# Uniform Laughlin gaps on large flat tori from finite Fock-space certificates

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**Abstract**

We develop a magnetic-periodization transfer of finite Fock-space gap certificates to the full contact interactions on a flat torus. In the normalization with planar nonzero pair energy one, the bosonic $V_0$ Hamiltonian obeys $H_B^2 \geq H_B/3$ whenever the shortest lattice vector is at least 20 in units of $\sqrt{2}\ell_B$. The fermionic $V_1$ Hamiltonian obeys $H_F^2 \geq \gamma H_F$ for every $\gamma < 0.04616733319001$ once that shortest vector is sufficiently large. Both estimates hold on the entire corresponding Fock space, uniformly in particle number and boundary twists. The geometric argument retains all center-of-mass theta modes, controls the failure of finite-torus pair projectivity relative to the interaction energy, and periodizes the connected three-body coefficient through an absolutely convergent COM partial trace. Positive finite four-body certificates survive without an inverse Gram matrix. Fermionic errors are controlled by Pauli bounds, while bosonic errors require coherent-state occupation and thermal-kernel comparisons with a positive four-body reserve. At flux $M = 2N$ or $M = 3N$, the bounds apply above the twofold or threefold Laughlin zero-mode space. The results concern large actual cycles and the full untruncated interactions; they do not assert a shape-uniform bound for arbitrary thin tori.

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# 1 Introduction

This manuscript is a candidate proof for independent review. Its mathematical conclusions and the narrower scope of selected machine checks are distinguished in Appendix D. The Laughlin wavefunctions provide explicit correlated states at fractional quantum Hall fillings, and contact pseudopotentials make them exact zero-energy states [6, 4]. A spectral-gap estimate asks a different question: can every orthogonal state be kept a positive energy above the entire zero-mode space as the particle number grows? On a torus this question also involves magnetic boundary conditions and a degenerate center-of-mass ground space. The familiar theta-function construction describes these zero modes [5, 3], but their existence alone supplies no uniform lower bound on the positive spectrum.

Finite Fock-space certificates provide a useful way to separate local algebra from geometry. The fermionic certificate of OpenAI [8] yields a uniform inequality for the full spherical interaction and its untruncated planar limit. Later work studies spherical stability [9], and Gao gives improved sphere-and-plane bounds through a semidefinite-program search [2]. We deliberately retain the earlier seven-row certificate rather than claim its constant is the strongest available. For bosons we use the explicit 138-row certificate in the candidate proof prepared and curated by Xin Shen [15]. Only its finite planar coefficient inequalities are inputs here; neither its proposed global theorem nor its partial formalization is assumed. Those finite inequalities, and the fermionic ones, admit direct independent rational verification.

The geometric distinction is essential. The OpenAI source concerns spherical fermionic $V_1$ and its planar limit; Shen's candidate concerns spherical bosonic $V_0$, with a proposed sufficient flux threshold $Q \geq 3088$ [8, 15]. Our proposed torus thresholds concern the actual shortest lattice cycle and use $M = 3N$ or $M = 2N$ at Laughlin filling, in place of the spherical shifted relations. Gao's spherical lower bound $1/5$ is an energy bound at fermionic filling $1/3$, and supplies no torus gap theorem [2].

The torus transfer is not a replacement of spherical $SU(2)$ averaging by a formal magnetic-translation average. A finite torus pair interaction is generally not a projector, raw periodized orbitals are not orthogonal, and a finite local approximation does not control all center-of-mass modes. A three-body single-contraction identity must respect its magnetic phases and the order of the COM trace. The bosonic problem adds unbounded occupation numbers: the bounded-annihilator argument available for fermions cannot be reused. These are the specific analytic steps addressed in this manuscript.

Our common construction periodizes the planar Gaussian jets without renormalizing them. Parameter Landau-level integration then preserves relative matrix units, while a relative-jet identity controls the Gram operator on the entire output theta space. Positive finite certificates survive by congruence even when their Gram matrices have kernels. The resulting full-Fock estimates are assembled by two distinct error budgets: fermionic errors are paid for by the pair energy, whereas bosonic coherent-state comparisons pay for them jointly with the pair energy and a positive four-body term. Theorems 2.1 and 2.2 state the resulting bounds.

The geometric condition is growth of the actual shortest noncontractible cycle at fixed magnetic length. Existing rigorous gap analyses for truncated thin-torus models [7, 11, 12] concern different Hamiltonians or limiting regimes and are not used as gap inputs for the full interactions here. Pairing and coherent-state descriptions of the torus [10, 1] inform the setting, but the all-COM frame estimate and the relative error bounds are proved below. All essential kernel, convergence, and occupation arguments are included in the manuscript. The finite exact datasets and executable checks accompany it separately from the finite-size numerical diagnostics.

# 2 Models and main statements

We consider the full lowest-Landau-level contact pseudopotential $V_0$ for bosons and $V_1$ for fermions on a flat magnetic torus. No hopping range, momentum sector, or COM sector is truncated. The energy convention is that the corresponding nonzero planar two-particle pseudopotential eigenvalue is one.

Finite-torus pair eigenvalues are not reset to one.

Let $T = \mathbb{C}/\Lambda$, $\mathrm{area}(\Lambda) = \pi M$, $M \in \mathbb{N}$, in the dimensionless coordinate $z = x/(\sqrt{2}\ell_B)$. Boundary twists are arbitrary and identical for all particles. The one-particle space $\mathcal{H}_1(T)$ has dimension $M$. The $N$-particle spaces are $\mathrm{Sym}^N \mathcal{H}_1(T)$ and $\bigwedge^N \mathcal{H}_1(T)$. The raw periodized orbitals $u_j^T(R)$ and the measure $\mathrm{d}\mu_k(R) = k\,\mathrm{d}^2R/\pi$ are fixed in Section 3. These orbitals retain their planar scale; they are neither normalized pointwise nor orthogonalized on the torus.

Write $a_j(R) = a(u_j^T(R))$ for bosonic annihilation and $c_j(R) = c(u_j^T(R))$ for fermionic annihilation. The COM-descendant pair annihilators are

$$B_p^B(R) = \frac{1}{\sqrt{2}} \sum_{i+j=p} \sqrt{\frac{p!}{2^p i!\, j!}}\, a_i(R) a_j(R), \tag{2.1}$$

$$v_p^{F,T}(R) = \sum_{\substack{i<j \\ i+j=p+1}} (i-j) \sqrt{\frac{p!}{2^p i!\, j!}}\, u_i^T(R) \wedge u_j^T(R), \qquad B_p^F(R) = A(v_p^{F,T}(R)). \tag{2.2}$$

In the bosonic case $v_p^{B,T} = B_p^{B\dagger}|\mathrm{vac}\rangle$. In particular, $B_0^B = a_0^2/\sqrt{2}$ and $v_0^{F,T} = -u_0^T \wedge u_1^T$. On the plane both $v_p$ families are orthonormal bases of the pair subspaces of relative degree zero and one, respectively. For $\sigma = B, F$ define

$$h_\sigma = \int_T |v_0^{\sigma,T}(R)\rangle\langle v_0^{\sigma,T}(R)|\, \mathrm{d}\mu_2(R), \qquad H_\sigma = \mathcal{L}_2(h_\sigma) = \int_T B_0^\sigma(R)^\dagger B_0^\sigma(R)\, \mathrm{d}\mu_2(R). \tag{2.3}$$

This is the periodized contact definition of the full interactions in the stated normalization. The positive lift $\mathcal{L}_2$ is defined below. All operators are bounded on each fixed $M, N$ sector. The bosonic full Fock space is an infinite direct sum, so its statements first hold on the finite-particle algebraic core and then as spectral inequalities for the direct-sum self-adjoint operator. In particular, $H^2 \geq \gamma H$ means $\|H\psi\|^2 \geq \gamma\langle\psi, H\psi\rangle$ on the form domain of $H^2$.

**Theorem 2.1** (Bosonic torus gap)**.** *For every lattice with $\ell(\Lambda) \geq 20$ and every boundary twist, the Hamiltonian (2.3) on the entire bosonic Fock space satisfies*

$$H_B^2 \geq \frac{1}{3} H_B. \tag{2.4}$$

*More generally, every $0 < \gamma < \gamma_B^*$ admits a systole threshold independent of particle number and twist, where*

$$\begin{aligned}\gamma_B^* = 1 &- \frac{76419170375082343297845 7}{150000000000000000000000} - \frac{19531250}{129140163} \\ &- 10^{-6}\left[12(9/8)^9 + 272(9/8)^{17}\right] \simeq 0.33724904107059384.\end{aligned} \tag{2.5}$$

*The length threshold 20 is dimensionless, equivalently $20\sqrt{2}\ell_B$ in physical units. It is a conservative sufficient condition, not an optimized finite-size estimate.*

**Theorem 2.2** (Fermionic torus gap)**.** *For every $0 < \gamma < \gamma_F^*$ there is a finite $L_\gamma$ such that, for every lattice with $\ell(\Lambda) \geq L_\gamma$ and every boundary twist, the full fermionic Fock-space Hamiltonian satisfies*

$$H_F^2 \geq \gamma H_F, \qquad \gamma_F^* = \frac{4616733319001}{10^{14}} = 0.04616733319001. \tag{2.6}$$

*The threshold is independent of particle number and twist. No explicit numerical value of $L_\gamma$ is asserted here.*

**Corollary 2.3** (Laughlin fillings ). *At $M = 2N$ for bosons and $M = 3N$ for fermions, the zero-energy spaces have dimensions two and three, respectively. The preceding inequalities bound the spectrum above these whole ground spaces. In any compact family of torus shapes at fixed magnetic length, they give thermodynamic gap bounds $1/3$ and $1/25$, respectively, for sufficiently large $N$. On the bosonic square sequence $M = 2N$, the explicit condition is already satisfied for $N \geq 64$.*

The kernel assertion is proved in Section 7. The fermionic estimate is assembled in Section 5 and the bosonic estimate in Section 6. Neither theorem includes a fixed short circumference or arbitrary degenerating thin-torus shapes. Moreover, on a sector with no zero mode, a full-Fock inequality excludes positive spectrum in $(0, \gamma)$; it does not assert a spacing $\gamma$ between the two lowest positive eigenvalues.

# 3 Magnetic periodization and local oscillator identities

Torus coherent states and their reproducing kernels provide the geometric background [1]. We specify the magnetic phases, measures, and raw-jet normalization below, and derive the full-COM frame estimate rather than assume it from a localized coherent-state construction.

## 3.1 Line bundles, measures, and the parameter Landau levels

Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \subset \mathbb{C}$, with $\mathrm{Im}(\overline{\omega}_1\omega_2) = \pi M$, and write $T = \mathbb{C}/\Lambda$. Lengths are measured in the coordinate $z = x/(\sqrt{2}\ell_B)$, where $\ell_B$ is the magnetic length. The systole is $\ell(\Lambda) = \min_{\lambda\in\Lambda\setminus\{0\}} |\lambda|$. All limits below concern this actual shortest lattice vector. Fix arbitrary real boundary angles $\theta_1, \theta_2$, and set

$$\chi(m\omega_1 + n\omega_2) = e^{i(m\theta_1+n\theta_2)}(-1)^{Mmn}, \qquad \chi_k = \chi^k. \tag{3.1}$$

For positive integer charge $k$, use the measure $\mathrm{d}\mu_k(z) = k\,\mathrm{d}^2z/\pi$ and magnetic translations

$$(T_x^{(k)}f)(z) = \exp\left[\frac{k}{2}(z\bar{x} - \bar{z}x)\right] f(z-x), \tag{3.2}$$

$$T_x^{(k)}T_y^{(k)} = e^{-ik\,\mathrm{Im}(x\bar{y})}T_{x+y}^{(k)}, \qquad \chi_k(\lambda)\chi_k(\mu) = e^{-ik\,\mathrm{Im}(\lambda\bar{\mu})}\chi_k(\lambda+\mu). \tag{3.3}$$

The charge-$k$ lowest Landau level (LLL) $\mathcal{H}_k(T)$ consists of Gaussian-holomorphic sections satisfying $T_\lambda^{(k)}f = \chi_k(\lambda)f$. Equivalently,

$$f(z+\lambda) = \chi_k(\lambda)^{-1}e^{k(z\bar{\lambda}-\bar{z}\lambda)/2}f(z). \tag{3.4}$$

Its dimension is $kM$. The plane reproducing kernel in the same measure is $K_k(z,w) = \exp[k(z\bar{w} - |z|^2/2 - |w|^2/2)]$. For a Gaussian times a polynomial define

$$P_kf = \sum_{\lambda\in\Lambda}\chi_k(\lambda)^{-1}T_\lambda^{(k)}f, \qquad K_k^T = P_kK_k. \tag{3.5}$$

These series and any fixed number of their derivatives converge absolutely. Reindexing gives the stated boundary law. Unfolding a fundamental-cell integral proves the reproducing property of $K_k^T$ with no additional area normalization. We do not assert that $P_k$ is a bounded map on the entire infinite-dimensional planar LLL. Its use is restricted to finite-degree vectors and the weighted, absolutely convergent series justified below.

Let $e_j(z) = z^je^{-|z|^2/2}/\sqrt{j!}$, $u_j(R) = T_R^{(1)}e_j$, and $u_j^T(R) = P_1u_j(R)$. A common-image unfolding gives, for finite Gaussian polynomials $f, g$,

$$\langle P_1f, P_1g\rangle_T = \sum_{\lambda\in\Lambda}\chi(\lambda)^{-1}\langle f, T_\lambda^{(1)}g\rangle_{\mathbb{C}}. \tag{3.6}$$

Indeed, write the two images as $\mu$ and $\mu+\lambda$. The cocycle and semicharacter factors cancel the common translation, and the translated cells tile the plane. Absolute Gaussian convergence permits this rearrangement. In particular, putting $b_\lambda(R)=\chi(\lambda)e^{i\,\mathrm{Im}(\lambda\bar{R})}$,

$$u_j^T(R+\lambda)=b_\lambda(R)u_j^T(R),\qquad g_{ij}^T(R,S)=\sum_{\lambda\in\Lambda}b_\lambda(S)^{-1}g_{ij}^{\mathrm{pl}}(R,S+\lambda),\tag{3.7}$$

where $g_{ij}(R,S)=\langle u_i(R),u_j(S)\rangle$. The phase in the second identity follows from $T_\lambda u_j(S)=e^{-i\,\mathrm{Im}(\lambda\bar{S})}u_j(S+\lambda)$.

For every fixed $R$, all $u_j^T(R)$ are physical one-particle LLL vectors in the variable $z$. Their dependence on the center $R$ is different: $u_j^T$ lies in the $j$th Landau level of the dual parameter line bundle. The annihilation amplitude is its conjugate and has the holomorphic-side automorphy, but is in the parameter LLL only for $j=0$. More generally, write $\mathcal{H}_{k,n}(T)$ for parameter charge $k$ and Landau level $n$, with $\mathcal{H}_k=\mathcal{H}_{k,0}$. A $k$-body center-of-mass (COM) vacuum seed has an amplitude in $\mathcal{H}_k$; its $n$th COM descendant has an amplitude in $\mathcal{H}_{k,n}$. A shared boundary multiplier alone is not the LLL differential condition.

## 3.2 The positive lift and exact COM integration

The positive normal-ordered lift is the Fock-space device used for the fermionic comparison in OpenAI [8, Section 2] and for the bosonic comparison in Shen [15, Section 1]. We retain its vacuum normalization. The parameter Landau-level identities that follow supply the torus replacement for the sources' spherical averaging step. In either statistics, let $\mathcal{L}_k(|v\rangle\langle w|)=A(v)^\dagger A(w)$, where $A(v)^\dagger|\mathrm{vac}\rangle=v$ with normalized symmetric or antisymmetric tensors. This linear map preserves positivity, equals the input operator in the $k$-particle sector, and vanishes below that sector. For bosonic occupation vectors, $A(\nu)=\prod_j a_j^{\nu_j}/\sqrt{\nu_j!}$; thus no extra $k!$ occurs. All few-body inequalities are lifted by this map, rather than estimated by counting particle subsets.

On dual parameter sections define

$$\mathfrak{a}_k=-\frac{\partial_R+k\bar{R}/2}{\sqrt{k}},\qquad \mathfrak{a}_k^\dagger=\frac{\partial_{\bar{R}}-kR/2}{\sqrt{k}},\qquad [\mathfrak{a}_k,\mathfrak{a}_k^\dagger]=1.\tag{3.8}$$

They preserve the boundary law and are adjoints in the cell integral. Differentiating the Gaussian series gives

$$\mathfrak{a}_1u_j^T=\sqrt{j}\,u_{j-1}^T,\qquad \mathfrak{a}_1^\dagger u_j^T=\sqrt{j+1}\,u_{j+1}^T.\tag{3.9}$$

These canonical commutation relations act on the infinite parameter Landau-level space, not inside the finite physical LLL.

**Lemma 3.1** (COM integration)**.** *Let $w,z$ be finite-degree $k$-body COM vacuum seeds of the chosen statistics. Define*

$$w_n^T=(\mathfrak{a}_k^\dagger)^n w^T/\sqrt{n!},\qquad z_m^T=(\mathfrak{a}_k^\dagger)^m z^T/\sqrt{m!}.\tag{3.10}$$

*Then*

$$\int_T|w_n^T(R)\rangle\langle z_m^T(R)|\,\mathrm{d}\mu_k(R)=\delta_{nm}\int_T|w^T(R)\rangle\langle z^T(R)|\,\mathrm{d}\mu_k(R).\tag{3.11}$$

*Different relative seeds need not be orthogonal; equal COM levels retain their full relative matrix unit.*

*Proof.* Evaluate the operator identity between fixed physical bras and kets. Integration by parts on the parameter line bundle has no boundary term, because the two boundary multipliers cancel. Move lowering operators across the integral, use $[\mathfrak{a}_k,(\mathfrak{a}_k^\dagger)^n]=n(\mathfrak{a}_k^\dagger)^{n-1}$, and annihilate the vacuum seeds. Unequal levels vanish; equal levels contribute $n!$, cancelled by the two normalizations. Polarization gives the identity for distinct seeds. □

Let $C_R e_j = u_j^T(R)$ and let $C_R^{(k)}$ denote its symmetric or exterior $k$th tensor action. If $X$ is an operator on the physical relative space, define

$$\Omega_k(X) = \int_T C_R^{(k)}(X \otimes |0_{\mathrm{COM}}\rangle\langle 0_{\mathrm{COM}}|) C_R^{(k)\dagger} \, \mathrm{d}\mu_k(R). \tag{3.12}$$

For finite support this is a positive map, including when the raw-jet Gram matrix has a kernel. For every finite-degree anchored operator $A$, Lemma 3.1 gives

$$\int_T C_R^{(k)} A C_R^{(k)\dagger} \, \mathrm{d}\mu_2(R) = \Omega_k\left(\frac{2}{k} \operatorname{Tr}_{\mathrm{COM}} A\right). \tag{3.13}$$

The corresponding plane twirl is

$$\int_{\mathbb{C}} T_R^{\otimes k} A T_R^{\otimes k\dagger} \, \mathrm{d}\mu_2(R) = \frac{2}{k} I_{\mathrm{COM}} \otimes \operatorname{Tr}_{\mathrm{COM}} A. \tag{3.14}$$

To verify the normalization, the normalized COM displacement is $h = \sqrt{k}R$. Writing $h = (q + ip)/\sqrt{2}$, Weyl conjugation sends the COM position kernel $A(x, y)$ to $e^{ip(x-y)}A(x - q, y - q)$. Integrating first in $p$ and then in $q$, with measure $\mathrm{d}q\,\mathrm{d}p/(2\pi)$, gives identity in COM and trace in that factor. The change of measure $\mathrm{d}\mu_2(R) = (2/k)\,\mathrm{d}^2h/\pi$ supplies the factor $2/k$. This argument also applies to relative-operator-valued kernels; it does not scalarize multiplicity spaces.

### 3.3 Uniform raw-jet bounds and the convergent domain

**Lemma 3.2** (Finite and high jets)**.** *For fixed $J$, define*

$$e_J(\Lambda) = \sup_R \|(\langle u_i^T(R), u_j^T(R)\rangle)_{0 \le i,j \le J} - I\|. \tag{3.15}$$

*There is a constant $C_J$, independent of flux, twist, particle number, and lattice shape, such that*

$$e_J(\Lambda) \le C_J \sum_{\lambda \neq 0} (1 + |\lambda|)^{2J} e^{-|\lambda|^2/2} \longrightarrow 0 \quad (\ell(\Lambda) \to \infty). \tag{3.16}$$

*For all lattices with $\ell(\Lambda) \ge 1$, a universal constant $C$ also satisfies*

$$\sup_R \|u_j^T(R)\|^2 \le C(1 + j), \qquad j \ge 0. \tag{3.17}$$

*Proof.* Equation (3.6) expresses the finite Gram matrix as its identity image plus oscillator displacement matrices. Each fixed matrix element is a Gaussian times a polynomial of degree at most $i + j$, proving (3.16) by a finite matrix norm bound. For the diagonal element use $\langle e_j, T_\lambda e_j\rangle = e^{-t/2}L_j(t)$, $t = |\lambda|^2$. Unitarity bounds its modulus by one, whereas the finite Laguerre series gives

$$|L_j(t)| \le \sum_{r=0}^{j} \frac{(jt)^r}{(r!)^2} \le e^{2\sqrt{jt}}. \tag{3.18}$$

For $t \ge 64j$, the product with $e^{-t/2}$ is at most $e^{-t/4}$. The points with $|\lambda| < 8\sqrt{j}$ number at most $C(1+j)$ by disjoint discs of radius $1/2$; the exterior Gaussian sum is uniformly bounded. The same packing argument bounds every fixed polynomial-Gaussian lattice sum uniformly in shape and makes it vanish as the systole grows. □

On the span of $e_0, \ldots, e_L$, the squared norm of $C_R$ is at most the sum of the squared column norms, hence at most $C(L + 1)^2$. Its $k$th tensor action has squared norm at most $C_k(L + 1)^{2k}$. Consequently all anchored degree blocks with trace norms bounded by a polynomial times $\rho^L$, $0 < \rho < 1$, remain absolutely summable after periodization. Equations (3.13) and (3.14) extend to precisely the exponentially weighted seeds used below by finite-degree truncation. This convergence is first established at fixed flux and particle sector; particle-number uniformity of the final gap comes from the relative energy bounds, not from a coarse many-body norm estimate.

### 3.4 A relative jet frame retaining every COM mode

**Lemma 3.3** (Full-COM relative frame)**.** *Let $a, b$ be positive integers, $t = a + b$, and $\nu = ab/t$. On $\mathcal{H}_a(T) \otimes \mathcal{H}_b(T)$ define the covariant normal jet along the diagonal by*

$$J_d F(C) = (\nu^d d!)^{-1/2}\, \partial_r^d \left\{ e^{\nu|r|^2/2} F(C + br/t, C - ar/t) \right\}\Big|_{r=0}. \tag{3.19}$$

*For a fixed finite set $\mathcal{D} \subset \{0, \dots, D\}$ the output is $\bigoplus_{d\in\mathcal{D}} \mathcal{H}_t(T)$, retaining all $tM$ COM modes in each summand. Its Gram operator satisfies the exact identity*

$$J_d J_e^\dagger = \sum_{\delta\in\Lambda} \chi_a(\delta)^{-1} M_{de}^{(\nu)}(\delta)\, T_{a\delta/t}^{(t)}, \qquad M_{de}^{(\nu)}(\delta) = \langle d|T_\delta^{(\nu)}|e\rangle. \tag{3.20}$$

*Every output translation in this formula is unitary on the same fixed-twist space. In particular,*

$$\|JJ^\dagger - I\| \le \epsilon_{a,b,D}(\Lambda) \le C_{a,b,D} \sum_{\delta\neq 0} (1 + |\delta|)^{2D} e^{-\nu|\delta|^2/2} \longrightarrow 0, \qquad J^\dagger J \le (1 + \epsilon_{a,b,D})I. \tag{3.21}$$

*Proof.* The weighted Jacobi change $C = (aR + bS)/t$, $r = R - S$ gives

$$a|R|^2 + b|S|^2 = t|C|^2 + \nu|r|^2, \qquad \mathrm{d}\mu_a(R)\, \mathrm{d}\mu_b(S) = \mathrm{d}\mu_t(C)\, \mathrm{d}\mu_\nu(r). \tag{3.22}$$

On the plane this is a unitary tensor factorization. In the normalized relative oscillator basis $e_d^{(\nu)}(r) = (\sqrt{\nu} r)^d e^{-\nu|r|^2/2}/\sqrt{d!}$, $J_d$ extracts the $d$th coefficient, so $J_d J_e^\dagger = \delta_{de} I$. This plane calculation defines the local diagonal jet on the torus; no global COM-times-relative torus factorization is assumed.

Expand both input reproducing kernels using (3.5). Under the Jacobi unitary $U$, their images obey

$$U(T_\lambda^{(a)} \otimes T_\mu^{(b)})U^\dagger = T_{(a\lambda+b\mu)/t}^{(t)} \otimes T_{\lambda-\mu}^{(\nu)}. \tag{3.23}$$

Set $\delta = \lambda - \mu$ and $\eta = a\delta/t$. There is no extra phase in this factorization: the magnetic bilinear form and Gaussian quadratic form split by the same change of variables. The remaining cocycles cancel explicitly:

$$\chi_a(\mu + \delta)^{-1} \chi_b(\mu)^{-1} = \chi_t(\mu)^{-1} \chi_a(\delta)^{-1} e^{-ia\,\mathrm{Im}(\mu\bar{\delta})}, \tag{3.24}$$

$$T_{\mu+\eta}^{(t)} = e^{ia\,\mathrm{Im}(\mu\bar{\delta})} T_\mu^{(t)} T_\eta^{(t)}. \tag{3.25}$$

Summing the common image $\mu$ recovers the charge-$t$ torus kernel. Taking relative oscillator matrix elements yields (3.20). For any $\gamma \in \Lambda$, the commutator of $T_\gamma^{(t)}$ and $T_\eta^{(t)}$ has phase $e^{-2ia\,\mathrm{Im}(\gamma\bar{\delta})} = 1$, since $\mathrm{Im}(\gamma\bar{\delta}) \in \pi M\mathbb{Z}$ and $aM$ is an integer. Thus the fractional lattice translation preserves the output boundary conditions.

The zero image is $\delta_{de} I$ on the entire output theta space. The oscillator formula bounds each nonzero image by

$$|M_{de}^{(\nu)}(\delta)| \le e^{-\nu|\delta|^2/2} \sqrt{d!e!} \sum_{l=0}^{\min(d,e)} \frac{(\sqrt{\nu}|\delta|)^{d+e-2l}}{l!(d-l)!(e-l)!}. \tag{3.26}$$

The triangle inequality and a finite matrix norm estimate prove (3.21). No factor $tM$ occurs. Tensoring with the identity of any spectator Hilbert space leaves the norm unchanged; this is the form used for Fock-valued annihilation amplitudes. □

### 3.5 The pair almost-projector and the common four-body frame

For relative degree $r = 0$ in the bosonic case and $r = 1$ in the fermionic case, define $Q\psi(R) = \langle v_0^T(R), \psi\rangle$. The target is all of $\mathcal{H}_2(T)$. On normalized symmetric tensors $Q = J_0$; on normalized antisymmetric tensors $Q = J_1$, because our seed is $v_0^F = -e_0 \wedge e_1 = (z_1 - z_2)/\sqrt{2}$ and the relative

charge is $1/2$. Thus neither case has an additional $\sqrt{2}$. Restriction to the appropriate statistics does not change $J_r J_r^\dagger$, since the relative jet already has that parity.

$$QQ^\dagger - I = \sum_{\delta\neq 0} \chi(\delta)^{-1} L_r(|\delta|^2/2) e^{-|\delta|^2/4} T^{(2)}_{\delta/2}, \qquad \epsilon_2^{(r)} = \sum_{\delta\neq 0} |L_r(|\delta|^2/2)| e^{-|\delta|^2/4}. \tag{3.27}$$

Here $L_0(x) = 1$ and $L_1(x) = 1 - x$. Since $h_T = Q^\dagger Q$, the relative estimate is the exact algebraic consequence

$$h_T^2 - h_T = Q^\dagger (QQ^\dagger - I) Q, \qquad -\epsilon_2 h_T \le h_T^2 - h_T \le \epsilon_2 h_T, \qquad \mathcal{L}_2(h_T^2) \ge (1-\epsilon_2) H. \tag{3.28}$$

For $\epsilon_2 < 1$, this also controls every nonzero pair eigenvalue in $[1-\epsilon_2, 1+\epsilon_2]$ and proves that $Q$ is onto. It cannot miss a high-COM small singular value.

Both pair families obey $v_p^T = (\mathfrak{a}_2^\dagger)^p v_0^T / \sqrt{p!}$. Hence Lemma 3.1 gives $\int_T B_p^\dagger B_q \, \mathrm{d}\mu_2 = \delta_{pq} H$. The amplitude $B_p$ has parameter level $p$, but $B_0$ is in the parameter LLL. For $F(R,S) = B_0(R) B_0(S) \psi$, apply Lemma 3.3 with $a = b = 2$. The relative ladder binomial expansion gives

$$J_d F = C_d \psi, \qquad C_d = \sum_{p=0}^{d} (-1)^{d-p} \sqrt{\binom{d}{p}} \, 2^{-d/2} B_p B_{d-p}, \qquad T_{4,D} = \int_T \sum_{d=0}^{D} C_d^\dagger C_d \, \mathrm{d}\mu_4. \tag{3.29}$$

Odd $d$ vanish because pair annihilators commute. The tensor-valued frame estimate yields

$$T_{4,D} \le (1 + \epsilon_{2,2,D}) S_4. \tag{3.30}$$

We use $D = 22$ for fermions and $D = 16$ for bosons. The fermionic proof converts the resulting error into a multiple of $H$ using bounded annihilators; the bosonic proof retains and pays for an $S_4$ contribution using occupation inequalities. These are different steps, and the latter never uses a bounded bosonic annihilator.

# 4 The exact three-body connected operator

The separation by contraction number is Wick normal ordering [13]. Its use to reduce the Laughlin gap to two-, three-, and four-body coefficients follows OpenAI [8, Section 2] and the bosonic adaptation in Shen [15, Section 1]. On a finite torus the double-contraction coefficient must be recomputed from the physical pair operator; it cannot be replaced by the spherical projector coefficient. Write $\sigma = B, F$ for bosons and fermions. Normal ordering the square of the full Hamiltonian gives

$$H_\sigma^2 = \mathcal{L}_2(h_\sigma^2) + S_3^\sigma + S_4^\sigma, \qquad S_4^\sigma = \int_{T^2} (B_0(R) B_0(S))^\dagger B_0(R) B_0(S) \, \mathrm{d}\mu_2(R) \, \mathrm{d}\mu_2(S). \tag{4.1}$$

The double-contraction term is $\mathcal{L}_2(h_\sigma^2)$, not $H_\sigma$. The zero-contraction term $S_4^\sigma$ is positive. The remaining term has one internal contraction. Its exact periodization requires both an unfolding identity and a COM partial trace.

## 4.1 The CCR and CAR kernels and their phases

For bosons set $a_R = a(u_0^T(R))$ and $g_{RS} = \langle u_0^T(R), u_0^T(S) \rangle$. The canonical commutation relations (CCR) give

$$a_R^2 a_S^{\dagger 2} = a_S^{\dagger 2} a_R^2 + 4 g_{RS} a_S^\dagger a_R + 2 g_{RS}^2. \tag{4.2}$$

Since $B_0(R) = a_R^2/\sqrt{2}$, the three-body integrand is $g_{RS} a_R^{\dagger 2} a_S^\dagger a_R a_S^2$. Define the unnormalized physical vectors $t_{RRS} = a_R^{\dagger 2} a_S^\dagger |\mathrm{vac}\rangle$ and $t_{RSS} = a_R^\dagger a_S^{\dagger 2} |\mathrm{vac}\rangle$. Its coefficient operator is therefore

$$\mathsf{S}_3^B = \int_{T^2} g_{RS} |t_{RRS}\rangle \langle t_{RSS}| \, \mathrm{d}\mu_2(R) \, \mathrm{d}\mu_2(S). \tag{4.3}$$

The double contraction is consistent with $\langle v_R, v_S\rangle = g_{RS}^2$: its Fock coefficient is $\frac{1}{2}g_{RS}^2 a_R^{\dagger 2}a_S^2$, exactly the kernel of $\mathcal{L}_2(h_B^2)$.

For fermions take $v_R = a \wedge b$, $v_S = c \wedge d$, $g_{xy} = \langle x, y\rangle$, and denote single-particle annihilation by $a(x)$. With $B_R = a(b)a(a)$ and $B_S = a(d)a(c)$, the canonical anticommutation relations (CAR) give

$$\begin{aligned} B_R B_S^\dagger = {} & \det(g) I - g_{ac}a(d)^\dagger a(b) + g_{bc}a(d)^\dagger a(a) \\ & + g_{ad}a(c)^\dagger a(b) - g_{bd}a(c)^\dagger a(a) + B_S^\dagger B_R. \end{aligned} \tag{4.4}$$

After multiplying by $B_R^\dagger$ on the left and $B_S$ on the right, the four single-contraction terms are

$$\begin{aligned} & - g_{ac}[a(d)B_R]^\dagger[a(b)B_S], \qquad + g_{bc}[a(d)B_R]^\dagger[a(a)B_S], \\ & + g_{ad}[a(c)B_R]^\dagger[a(b)B_S], \qquad - g_{bd}[a(c)B_R]^\dagger[a(a)B_S]. \end{aligned} \tag{4.5}$$

Integrating them gives $S_3^F$. This calculation fixes the signs without assuming that the moving torus jets are orthogonal.

In each of these bosonic or fermionic coefficient kernels, the exterior rank-one operator $E(R, S)$ contains one $S$-centered orbital in the ket and two in the bra. Equation (3.7) thus gives

$$E(R, S+\lambda) = b_\lambda(S)^{-1}E(R, S), \qquad g_{ij}^T(R, S)E(R, S) = \sum_{\lambda\in\Lambda} g_{ij}^{\mathrm{pl}}(R, S+\lambda)E(R, S+\lambda). \tag{4.6}$$

The phases agree term by term, not merely in modulus. The sum and the $S$-cell integral therefore unfold to $S \in \mathbb{C}$, while $R$ remains in a cell and every exterior orbital remains periodized. Fixed-order planar contractions are polynomial-Gaussian, and exterior jets change only by unit phases under cell translations, so the unfolding is absolutely convergent. One internal edge leaves no unaccounted winding variable.

## 4.2 Anchored operators and the order of the limiting operations

On the planar three-particle space put $A_B = \frac{1}{2}n_0(n_0 - 1)$ and $A_F = n_0 n_1$, and define

$$K_\sigma = A_\sigma(H_{3,\sigma}^{\mathrm{pl}} - I). \tag{4.7}$$

The unfolded kernel is $\int_T C_R^{(3)} K_\sigma C_R^{(3)\dagger}\, \mathrm{d}\mu_2(R)$. To see the subtraction in (4.7), normal order $A_\sigma H_{3,\sigma}^{\mathrm{pl}}$ at the fixed first center. The four-body term vanishes on three particles, and the integrated double contraction is $A_\sigma$, since the planar pair interaction is the projector onto the pair space. The single contraction is what remains. For bosons one can also read the seed directly as

$$K_B = \int_{\mathbb{C}} e^{-|\beta|^2/2}|t_{00\beta}\rangle\langle t_{0\beta\beta}|\, \mathrm{d}\mu_2(\beta). \tag{4.8}$$

For example, with $E(0, \beta) = |t_{00\beta}\rangle\langle t_{0\beta\beta}|$, the precise covariance is

$$\begin{aligned} C_R^{(3)}E(0, \beta)C_R^{(3)\dagger} &= e^{i\,\mathrm{Im}(R\bar\beta)}E_T(R, R+\beta), \\ g_{00}^{\mathrm{pl}}(R, R+\beta) &= e^{-|\beta|^2/2+i\,\mathrm{Im}(R\bar\beta)}. \end{aligned} \tag{4.9}$$

Translation covariance supplies the same phase in this seed and the contraction kernel. Neither anchored operator is generally self-adjoint, but its integral is; replacing it by its Hermitian part has no effect.

The required infinite sums have explicit bounds. In the bosonic total-degree $l > 0$ block, $A_B$ has support only on $|0, 0, l\rangle$, and

$$(H_{3,B}^{\mathrm{pl}} - I)|0, 0, l\rangle = 2\,2^{-l/2}B_l^\dagger|e_0\rangle_{1p}, \qquad \|K_{B,l}\|_1 \le 2\sqrt{3}\,2^{-l/2}. \tag{4.10}$$

The degree-zero scalar is 6. The vector on the right is created from the one-particle $e_0$ state, not the vacuum; its norm is bounded by $\sqrt{3}$ because pair insertion into three bosons is $\sqrt{3}$ times a symmetric projection. In the fermionic degree $L = l + 1$ block, $l \geq 2$,

$$K_{F,L} = |0,1,l\rangle\langle 0,1,l|(H^{\mathrm{pl}}_{3,F} - I),$$
$$\|K_{F,L}\|_1 \leq [\sqrt{2l} + l - 1]2^{-l/2}. \tag{4.11}$$

Here the 01 pair contributes exactly the subtracted identity. The other two pair coefficients have magnitudes $\sqrt{l}\,2^{-(l-1)/2}$ and $(l-1)2^{-l/2}$; wedging a normalized pair with a normalized orbital has norm at most one. Each block has rank at most one, so these operator-norm bounds are also trace-norm bounds.

The legitimate order is: truncate total degree; perform the unitary relative-COM decomposition in each finite block; periodize its seeds and descendants; integrate the parameter ladders; then remove the degree cutoff. Equal descendant levels retain all relative matrix units, including off-diagonal ones. Lemma 3.2 bounds the squared threefold periodization norm by $C(L+1)^6$, which is summable against (4.10) or (4.11). Thus

$$\mathsf{S}^{\sigma}_3 = \Omega_3\left(\frac{2}{3}\,\mathrm{Tr}_{\mathrm{COM}}\,K_\sigma\right). \tag{4.12}$$

An integrated planar spectral formula cannot replace $K_\sigma$ pointwise. For example, the fermionic anchored operator has a nonzero matrix element on $|0,1,3\rangle$, whereas the relative highest seed of the corresponding total degree is zero. COM descendants are removed only by the integral, not beforehand.

### 4.3 Relative eigenvalues from compressed exchanges

Compressed pair exchanges are the recoupling mechanism behind the fermionic three-body spectrum in OpenAI [8, Section 3] and its bosonic counterpart in Shen [15, Section 1]. Here we derive the planar oscillator coefficients in the normalization needed by the torus transfer. The anchored unfolding and the convergence argument above are additional geometric obligations; a planar spectral formula alone does not establish them. Let $U$ be the planar one-particle LLL and $V_\sigma$ its relative-degree-zero bosonic or relative-degree-one fermionic pair space. Define pair insertion $W^B_3(v\otimes u) = B(v)^\dagger u$ and $W^F_3(v\otimes u) = v\wedge u$. Let $\iota : V_\sigma \otimes U \to U^{\otimes 3}$ be the normalized tensor embedding, with symmetry or antisymmetry in the first pair, and $S_{ij} = \iota^\dagger \tau_{ij}\iota$ the compression of the ambient permutation. The bare permutation need not preserve the domain. Then

$$W^{B\dagger}_3 W^B_3 = I + S_{13} + S_{23}, \qquad W^{F\dagger}_3 W^F_3 = I - S_{13} - S_{23}, \qquad W^\sigma_3 W^{\sigma\dagger}_3 = H^{\mathrm{pl}}_{3,\sigma}. \tag{4.13}$$

The pair COM and spectator oscillator have one relative oscillator after separating their total COM. Normalized relative highest seeds are

$$\xi^\sigma_z = \sum_{p=0}^{z} s^\sigma_{zp} v^\sigma_p \otimes e_{z-p}, \qquad w^\sigma_z = W^\sigma_3 \xi^\sigma_z,$$
$$s^B_{zp} = (-1)^p\sqrt{\binom{z}{p}}\,3^{-p/2}(2/3)^{(z-p)/2}, \qquad s^F_{zp} = (-1)^{z-p}\sqrt{\binom{z}{p}}\,3^{-p/2}(2/3)^{(z-p)/2}. \tag{4.14}$$

Each compressed exchange is scalar on this one-dimensional relative level, since it intertwines COM translations and preserves total degree. For bosons, its $v_0 \otimes e_z$ component receives only the $p = z$ contribution. The ratio of seed coefficients is $(-1)^z 2^{-z/2}$ and the ordered pair tensor coefficient is $2^{-z/2}$, giving $b^B_z = (-1/2)^z$. For fermions the two contributions are $p = z-1, z$. With antisymmetric continuation of the planar pair coefficients,

$$b^F_z = \frac{-s^F_{z,z-1}v^F_{z-1}(0,z) + s^F_{z,z}v^F_z(1,z)}{2s^F_{z,0}} = (-1)^{z-1}\frac{3z-1}{2^{z+1}}. \tag{4.15}$$

Out-of-range terms vanish; the expression also gives $z = 0, 1$ directly. Therefore $W_3^{\sigma\dagger} W_3^{\sigma}$ has eigenvalue $1 + q_z^{\sigma}$, where

$$q_z^B = 2(-1/2)^z, \qquad q_z^F = (3z-1)(-1/2)^z. \tag{4.16}$$

Using (4.13),

$$(H_{3,\sigma}^{\mathrm{pl}})^2 - H_{3,\sigma}^{\mathrm{pl}} = I_{\mathrm{COM}} \otimes \sum_{z\geq 0} q_z^{\sigma} |w_z^{\sigma}\rangle\langle w_z^{\sigma}|. \tag{4.17}$$

On the other hand, $\int A_\sigma(R)\,\mathrm{d}\mu_2(R) = H_{3,\sigma}^{\mathrm{pl}}$ and the latter commutes with simultaneous translations. Equation (3.14) applied to (4.7) identifies $(2/3)\,\mathrm{Tr}_{\mathrm{COM}}\,K_\sigma$ with this relative coefficient. Combining with (4.12) proves the following proposition.

**Proposition 4.1** (Exact connected coefficients)**.** *Let $D_z^{\sigma}(R) = A(w_z^{\sigma,T}(R))$. In either statistics, on each finite particle sector,*

$$S_3^B = \int_T \sum_{z\geq 0} 2(-1/2)^z D_z^{B\dagger} D_z^B \,\mathrm{d}\mu_3, \tag{4.18}$$

$$S_3^F = \int_T \sum_{z\geq 0} (3z-1)(-1/2)^z D_z^{F\dagger} D_z^F \,\mathrm{d}\mu_3. \tag{4.19}$$

*The series are defined by the absolutely convergent anchored construction above. For bosons $w_1^B = 0$; for fermions $w_0^F = w_1^F = w_3^F = 0$. These are algebraic zero vectors because $\|w_z^{\sigma}\|^2 = 1 + q_z^{\sigma} = 0$, so periodization preserves their vanishing. All other negative odd-degree terms must be controlled, rather than discarded.*

# 5 The fermionic comparison and its uniform error budget

We now complete the proof of Theorem 2.2. Throughout this section, $H = H_{\mathrm{F}}$ is the full $V_1$ Hamiltonian with the normalization fixed in the model definition. We abbreviate $B_p = B_p^F$, $D_z = D_z^F$, $C_3 = S_3^F$, $S_4 = S_4^F$, $h_T = h_F$, and $\varepsilon_2^{\mathrm{F}} = \epsilon_2^{(1)}$. We use the raw periodized jets $c_j(R) = c(u_j^T(R))$. The geometric estimates and the exact center-of-mass (COM) integration identity were established in Lemmas 3.2, 3.3, and 3.1. The remaining task is to compare the three- and four-body terms with a fixed sum of squares. All errors will be bounded by a scalar multiple of $H$.

The finite coefficient data are those of the Fock-space certificate in [8]. Their role here is entirely algebraic: Appendix A prints all seven integer rows and gives the rational calculations proving the finite inequalities used below. No spherical spectral-gap statement is an input to this argument.

## 5.1 A four-body frame error paid for by the pair energy

The zero-contraction term in the normal-ordered square of $H$ is

$$S_4 = \int_{T\times T} (B_0(R)B_0(S))^{\dagger}(B_0(R)B_0(S))\;\mathrm{d}\mu_2(R)\,\mathrm{d}\mu_2(S) \geq 0. \tag{5.1}$$

For a Fock-space vector $\psi$, the function $B_0(R)B_0(S)\psi$ lies in $\mathcal{H}_2(T)\otimes\mathcal{H}_2(T)$ with values in the remaining Fock space. Apply the finite relative-jet map with charges $a = b = 2$ and relative degrees $0 \leq d \leq 22$. Its components are

$$C_d(R) = \sum_{p=0}^{d} (-1)^{d-p} \sqrt{\binom{d}{p}}\, 2^{-d/2} B_p(R) B_{d-p}(R), \qquad T_4 = \int_T \sum_{d=0}^{22} C_d(R)^{\dagger} C_d(R)\;\mathrm{d}\mu_4(R). \tag{5.2}$$

Pair annihilators commute because each has even fermionic degree. Consequently $C_d = 0$ for odd $d$, by exchanging $p$ and $d - p$. Lemma 3.3, applied to its tensor product with the identity on the spectator space, gives

$$T_4 \le (1 + \varepsilon_4^{\mathrm{F}}(\Lambda))S_4, \qquad \varepsilon_4^{\mathrm{F}}(\Lambda) = \epsilon_{2,2,22}(\Lambda) \xrightarrow[\ell(\Lambda)\to\infty]{} 0. \tag{5.3}$$

This use of the frame bound keeps every COM theta mode and does not introduce the dimension of the spectator space. A second estimate converts its error into pair energy.

The method has a spherical antecedent in the fixed-block relative-energy estimate of OpenAI [9, Section 7, vanishing-weight lemma and fixed-block-energy estimate]: first bound the finite left pair annihilator, then average the remaining pair energy. The torus proof below supplies its raw-Gram correction and uses the exact parameter pair integral.

**Lemma 5.1** (Finite left-factor estimate)**.** *For $0 \le p \le 22$, and for every center $R$,*

$$\|B_p(R)\|^2 \le (p+2)(1 + e_{23}(\Lambda))^2. \tag{5.4}$$

*In particular,*

$$T_4 \le C_4^{\mathrm{F}}(\Lambda)H, \qquad C_4^{\mathrm{F}}(\Lambda) = 5152(1 + e_{23}(\Lambda))^2. \tag{5.5}$$

*Proof.* Let $C_R : e_j \mapsto u_j^T(R)$ be restricted to the first $p + 2$ orbitals. Its Gram matrix is bounded above by $(1 + e_{23})I$. The planar pair vector $v_p$ has norm one, so the periodized vector $(\bigwedge^2 C_R)v_p$ has norm at most $1 + e_{23}$. A two-form on a space of dimension at most $p + 2$ has a canonical orthogonal decomposition $\sum_{a=1}^{r} \sigma_a f_{2a-1} \wedge f_{2a}$, where $r \le (p + 2)/2$ and $\sum_a |\sigma_a|^2$ is its squared norm. The corresponding annihilator has norm at most $\sum_a |\sigma_a|$, since each product of two normalized one-particle annihilators is a contraction. Cauchy–Schwarz therefore bounds its squared norm by $r \sum_a |\sigma_a|^2$, which implies the slightly weaker bound (5.4).

The coefficients in (5.2) have squared sum one. Thus, for every $\psi$,

$$\begin{aligned} \|C_d(R)\psi\|^2 &\le \sum_{p=0}^{d} \|B_p(R)B_{d-p}(R)\psi\|^2 \\ &\le (1 + e_{23})^2 \sum_{p=0}^{d} (p+2)\|B_{d-p}(R)\psi\|^2. \end{aligned} \tag{5.6}$$

Integrating, use $\mathrm{d}\mu_4 = 2\,\mathrm{d}\mu_2$ and the exact identity $\int_T B_q^\dagger B_q \,\mathrm{d}\mu_2 = H$ for every $q$, from Lemma 3.1. This gives

$$T_4 \le 2(1 + e_{23})^2 \sum_{d=0}^{22} \sum_{p=0}^{d} (p+2)H = 5152(1 + e_{23})^2 H. \tag{5.7}$$

Every estimate used a bounded left factor followed by an exact pair-energy integral, so the result is independent of particle number. □

Combining (5.3) and (5.5) yields the form that will enter the final budget:

$$S_4 \ge \frac{T_4}{1 + \varepsilon_4^{\mathrm{F}}} \ge T_4 - C_4^{\mathrm{F}}(\Lambda)\varepsilon_4^{\mathrm{F}}(\Lambda)H. \tag{5.8}$$

The last inequality follows from $1 - (1 + x)^{-1} = x/(1 + x) \le x$ for $x \ge 0$. In particular, the error does not multiply $S_4$, a particle-counting operator, or an identity with a size-dependent coefficient.

## 5.2 Pauli control of the complete negative three-body tail

The exact three-body identity (4.19) contains all relative degrees. Its negative tail cannot be controlled by extending a fixed-degree Gram estimate to arbitrarily high jets. The high-jet estimate of Lemma 3.2, $\sup_R \|u_j^T(R)\|^2 \le C(1+j)$ for $\ell(\Lambda) \ge 1$, supplies the uniform majorant needed for the following Pauli estimate.

Recall the coefficients and annihilators in the exact three-body identity:

$$\begin{aligned} q_z &= (3z-1)(-1/2)^z, \qquad D_z(R) = \sum_{p=0}^{z} s_{z,p}\, c_{z-p}(R) B_p(R), \\ s_{z,p} &= (-1)^{z-p} \sqrt{\binom{z}{p}}\, 3^{-p/2} (2/3)^{(z-p)/2}. \end{aligned} \tag{5.9}$$

The coefficients have squared sum one, and $D_0, D_1, D_3$ vanish as algebraic wedge identities. All even terms are nonnegative. Define

$$M_z(\Lambda) = \sup_{R \in T,\, 0 \le j \le z} \|u_j^T(R)\|^2, \qquad \delta_\Lambda = \frac{3}{2} \sum_{\substack{z \ge 17 \\ z \text{ odd}}} |q_z| (z+1) M_z(\Lambda). \tag{5.10}$$

The canonical anticommutation relations give $\|c_j(R)\| = \|u_j^T(R)\|$. Cauchy–Schwarz in the coefficient index therefore gives

$$\|D_z(R)\psi\|^2 \le \sum_{p=0}^{z} \|c_{z-p}(R) B_p(R)\psi\|^2 \le M_z(\Lambda) \sum_{p=0}^{z} \|B_p(R)\psi\|^2. \tag{5.11}$$

After integrating with $\mathrm{d}\mu_3 = (3/2)\,\mathrm{d}\mu_2$, the exact pair-frame identity pays for each of the $z+1$ summands by $H$. Thus the negative tail satisfies

$$\int_T \sum_{\substack{z \ge 17 \\ z \text{ odd}}} q_z D_z^\dagger D_z \, \mathrm{d}\mu_3 \ge -\delta_\Lambda H. \tag{5.12}$$

For each fixed $z$, $M_z(\Lambda) \to 1$. Lemma 3.2 gives $M_z \le C(1+z)$ uniformly for $\ell \ge 1$, while $|q_z|$ decays geometrically. Dominated convergence therefore yields

$$\lim_{\ell(\Lambda)\to\infty} \delta_\Lambda = \frac{3}{2} \sum_{\substack{z \ge 17 \\ z \text{ odd}}} (3z-1)(z+1) 2^{-z} = \sum_{m=8}^{\infty} (9m^2 + 12m + 3) 4^{-m} = \frac{61}{4096}. \tag{5.13}$$

The middle expression uses $z = 2m+1$; the last equality follows by twice differentiating the geometric series beginning at $m = 8$. Both the convergence and the resulting constant are uniform in particle number, boundary twist, and lattice shape subject to the stated systole condition.

## 5.3 Decoding the seven rows and the finite coupled spaces

Set $P = 10^7$. For row $\rho$, the printed integers are $t_\rho$, $\ell_\rho$, and $a_{pj}^\rho$; missing entries are zero. Define

$$\lambda_\rho = \frac{\ell_\rho}{P}, \qquad \alpha_{pj}^\rho = \frac{a_{pj}^\rho}{P} \sqrt{\frac{2^{p-t_\rho} t_\rho! (p+j-t_\rho)!}{p! j!}}, \qquad \eta = \sum_{\rho=1}^{7} \lambda_\rho^2 = \frac{93527408868499}{10^{14}}. \tag{5.14}$$

The seven $t$ values are $0, 1, 1, 2, 5, 5, 7$. Every nonzero entry satisfies $0 \le p \le 7$, $0 \le j \le 8$, and $0 \le p + j - t_\rho \le 8$. These support conditions bound all auxiliary one-particle modes by eight, although their coupled four-body coefficients require jets up to degree twenty-four.

To specify the coefficients without referring to a spherical limit, take $0 < u < 1$, put $v = \sqrt{1-u^2}$, and define $s^{(u)}_{z,T,p}$ by the exact polynomial identity

$$\frac{(uX - vY)^z (vX + uY)^{T-z}}{\sqrt{z!(T-z)!}} = \sum_{p=0}^{T} s^{(u)}_{z,T,p} \frac{X^p Y^{T-p}}{\sqrt{p!(T-p)!}}. \tag{5.15}$$

A coefficient is zero if any of its degree indices is outside its indicated range. The change of variables is orthogonal and preserves the normalized-monomial inner product. For a fixed $T$, the rows indexed by $z$ consequently form an orthogonal matrix. The factor $(vX + uY)^{T-z}/\sqrt{(T-z)!}$ is the normalized COM descendant. This identifies the algebraic transformation with the ladder normalization of Lemma 3.1.

For the three-body comparison, set $u = 1/\sqrt{3}$ and abbreviate $s = s^{(1/\sqrt{3})}$. Define

$$e_z = \sum_{\rho=1}^{7} \sum_{T=\max(z,t_\rho)}^{15} \left[ 2\lambda_\rho s_{z,T,t_\rho} \sum_{p+j=T} \alpha^\rho_{pj} s_{z,T,p} + \left( \sum_{p+j=T} \alpha^\rho_{pj} s_{z,T,p} \right)^2 \right]. \tag{5.16}$$

The exact rational calculation in Appendix A.2 proves the thirteen inequalities

$$e_z < \frac{3}{2} q_z, \qquad z \in \{2, 4, 5, \dots, 15\}. \tag{5.17}$$

The excluded indices $0, 1, 3$ correspond to identically zero wedge images and require no scalar comparison.

For four bodies, let $\mathcal{R}_D = \{1, 3, 5, \dots\} \cap [1, D]$, where $1 \le D \le 23$. In the formal orthonormal input basis $v_p \otimes (e_j \wedge e_k)$, $j < k$, define

$$S_r(D, T; p, j, k) = \sqrt{2}\, s^{(1/\sqrt{2})}_{r,j+k,j} s^{(1/\sqrt{2})}_{D-r,T-r,p}. \tag{5.18}$$

This is zero if $r > j+k$ and is extended antisymmetrically in $j, k$. The first coupling separates the relative degree $r$ of the two single-particle factors; odd $r$ is precisely their antisymmetry condition. The second coupling separates their pair COM from that of $v_p$. Its relative degree is $D - r$, and $T - D$ is the total COM descendant degree. The highest-weight wedge columns are

$$w_{D,r} = \sum_{\substack{p+j+k=D \\ j<k}} S_r(D, D; p, j, k)\, v_p \wedge e_j \wedge e_k, \qquad W_D = (w_{D,r})_{r \in \mathcal{R}_D}, \qquad G_D = W_D^\dagger W_D. \tag{5.19}$$

Each column has physical orbital degree $D + 1$, because $v_p$ has degree $p + 1$. With $i = p + j - t_\rho$, $k = q + l - t_\rho$, and $T = p + j + k = q + l + i$, set

$$(E_D)_{rs} = -\sum_{\rho=1}^{7} \sum_{\substack{(p,j),(q,l) \in \mathrm{supp}(a^\rho) \\ T \ge D}} \alpha^\rho_{pj} \alpha^\rho_{ql} S_r(D, T; p, j, k) S_s(D, T; q, l, i). \tag{5.20}$$

The entry pairs are ordered; exchanging them proves that $E_D$ is real symmetric. Their support implies $T \le 23$. Let $\Pi_1$ denote the projection onto the copy index $r = 1$, and fix $\varepsilon_{\mathrm{F}} = 3 \times 10^{-6}$. The four-body certificate is

$$G_D(2\Pi_1 - E_D + \varepsilon_{\mathrm{F}} I) G_D \ge 0, \qquad 1 \le D \le 23. \tag{5.21}$$

Here $\Pi_1$ has no relation to the boundary semicharacter $\chi$. Appendix A.3 gives a radical-free rational reconstruction of every matrix in (5.21), together with an exact positivity test.

### 5.4 Normal ordering of the positive comparison operator

Using the fixed coefficients above, define the torus operators

$$F_\rho(R) = \lambda_\rho B_{t_\rho}(R) + \sum_{p,j} \alpha_{pj}^\rho c_{p+j-t_\rho}(R)^\dagger c_j(R) B_p(R), \qquad K_T = \int_T \sum_{\rho=1}^{7} F_\rho(R)^\dagger F_\rho(R)\, \mathrm{d}\mu_2(R) \geq 0. \tag{5.22}$$

Every term ends in a pair annihilator. In particular, if $H\psi = 0$, the exact pair-frame identity implies $B_p(R)\psi = 0$ for every $p$ and almost every $R$, and continuity then gives this equality for all $R$. Hence $F_\rho(R)\psi = 0$. The comparison thus preserves the zero-mode ideal exactly.

For one row, suppress $\rho$ and $R$, and put $i = p + j - t$, $k = q + l - t$. Write $g_{ik} = \langle u_i^T, u_k^T \rangle$ and $O_{p,j,k} = c_k c_j B_p$. Applying $c_i c_k^\dagger = g_{ik} - c_k^\dagger c_i$ yields the exact decomposition

$$(F^\dagger F)_2 = \lambda^2 B_t^\dagger B_t, \tag{5.23}$$

$$(F^\dagger F)_3 = \sum_{p,j} \lambda \alpha_{pj} \big[ (c_i B_t)^\dagger c_j B_p + (c_j B_p)^\dagger c_i B_t \big] + \sum_{p,j;q,l} \alpha_{pj}\alpha_{ql} g_{ik} (c_j B_p)^\dagger c_l B_q, \tag{5.24}$$

$$(F^\dagger F)_4 = -\sum_{p,j;q,l} \alpha_{pj}\alpha_{ql} O_{p,j,k}^\dagger O_{q,l,i}. \tag{5.25}$$

Only the single contraction in (5.24) contains the local Gram matrix. Let $K_3^{(0)}$ be its integral with $g_{ik}$ replaced by $\delta_{ik}$, including the displayed cross terms, and let $R_{3,\mathrm{metric}}$ be the difference. Let $K_4$ be the integral of (5.25), summed over rows. The pair-frame identity then gives

$$K_T = \eta H + K_3^{(0)} + R_{3,\mathrm{metric}} + K_4. \tag{5.26}$$

We identify $K_3^{(0)}$ explicitly. Before wedging, its coefficient matrix is on the formal input space $v_p \otimes e_j$. The Kronecker condition $i = k$ says exactly that $p + j = q + l = T$. In a cross term the second input is $v_t \otimes e_{T-t}$, so total degree is again preserved. At fixed $T$, change to the orthonormal coupled basis (5.15). A relative index $z$ has COM descendant degree $T - z$. The integral in Lemma 3.1 kills matrix units with unequal descendant degrees and replaces each equal-descendant diagonal block by its highest-weight matrix unit. This traces the COM factor but does not delete any relative matrix entries. The diagonal coefficient that remains is exactly (5.16). Hence

$$K_3^{(0)} = \int_T \sum_{z=0}^{15} e_z D_z(R)^\dagger D_z(R)\, \mathrm{d}\mu_2(R). \tag{5.27}$$

Using (5.17), $\mathrm{d}\mu_3 = (3/2)\,\mathrm{d}\mu_2$, the three zero wedge identities, and the nonnegativity of even degrees $z \geq 16$, the exact three-body term satisfies

$$C_3 \geq K_3^{(0)} - \delta_\Lambda H. \tag{5.28}$$

**Lemma 5.2** (The finite Gram error is relative to $H$)**.** *Let $n_\rho$ be the number of nonzero entries in row $\rho$, and define*

$$C_\alpha^{\mathrm{F}} = \sum_{\rho=1}^{7} n_\rho \max_{0 \leq i \leq 8} \sum_{p+j-t_\rho = i} (\alpha_{pj}^\rho)^2 = \frac{297842353828763087}{16800000000000000}. \tag{5.29}$$

*Then, as quadratic-form inequalities on the fermionic Fock space,*

$$-C_\alpha^{\mathrm{F}} e_8(1+e_8) H \leq R_{3,\mathrm{metric}} \leq C_\alpha^{\mathrm{F}} e_8(1+e_8) H. \tag{5.30}$$

*Proof.* For a fixed row and center, put $X_i = \sum_{p+j-t_\rho=i} \alpha^\rho_{pj} c_j B_p$. The metric remainder is $\sum_{i,k}(g_{ik} - \delta_{ik})X_i^\dagger X_k$. Since $\|g - I\| \le e_8$, its absolute quadratic form on $\psi$ is at most $e_8 \sum_i \|X_i\psi\|^2$. Cauchy–Schwarz gives

$$\begin{aligned}\sum_i \|X_i\psi\|^2 &\le \left(\max_i \sum_{p+j-t_\rho=i} (\alpha^\rho_{pj})^2\right) \sum_{(p,j)\in\mathrm{supp}(a^\rho)} \|c_j B_p \psi\|^2 \\ &\le (1+e_8)\left(\max_i \sum_{p+j-t_\rho=i} (\alpha^\rho_{pj})^2\right) \sum_{(p,j)\in\mathrm{supp}(a^\rho)} \|B_p\psi\|^2. \end{aligned} \tag{5.31}$$

The second inequality uses $\|c_j\|^2 = g_{jj} \le 1 + e_8$. Integrating each term with $\mathrm{d}\mu_2$ gives $\langle\psi, H\psi\rangle$ exactly; there are $n_\rho$ terms. Summing over rows proves (5.30). Finally, squaring (5.14) removes every radical, so the value in (5.29) is computed using rational arithmetic alone. □

## 5.5 Kernel-preserving four-body transfer and the count 1222

The four-body term (5.25) has no contraction and therefore no local Gram correction. Change each fixed-$T$ formal input block to the coupled basis (5.18). Integrating COM descendants by Lemma 3.1 leaves its copy matrix $E_D$. If $w^T_{D,r}(R) = (\bigwedge^4 C_R) w_{D,r}$, this gives the exact identity

$$\begin{aligned} K_4 &= \int_T \sum_{D=1}^{23} \sum_{r,s\in\mathcal{R}_D} (E_D)_{rs} A(w^T_{D,r}(R))^\dagger A(w^T_{D,s}(R))\ \mathrm{d}\mu_2(R) \\ &= \frac{1}{2}\int_T \sum_{D=1}^{23} \sum_{r,s\in\mathcal{R}_D} (E_D)_{rs} A(w^T_{D,r}(R))^\dagger A(w^T_{D,s}(R))\ \mathrm{d}\mu_4(R). \end{aligned} \tag{5.32}$$

At copy index $r = 1$, the second two-particle factor also has relative degree one. The remaining coupling has relative degree $d = D-1$, and its coefficient is the normalized binomial coefficient in (5.2). Therefore

$$A(w^T_{D,1}(R)) = C_{D-1}(R), \qquad T_4 = 2\int_T \sum_{D=1}^{23} A(w^T_{D,1}(R))^\dagger A(w^T_{D,1}(R))\ \mathrm{d}\mu_2(R). \tag{5.33}$$

Both $\mathrm{d}\mu_4 = 2\,\mathrm{d}\mu_2$ and $D = d+1$ are needed here; together they produce $2\Pi_1$ in the certificate. For even $D$, the $r = 1$ column is antisymmetric under exchange of its two pairs, while wedging two degree-two vectors is symmetric, so that column vanishes. The formulas do not require this column, or any other column, to be nonzero or independent.

To transfer the certificate, let $A_D = 2\Pi_1 - E_D + \varepsilon_{\mathrm{F}} I$. The inequalities $G_D A_D G_D \ge 0$ and $W_D A_D W_D^\dagger \ge 0$ are equivalent: both test the same quadratic form on $\mathrm{ran}(G_D) = \mathrm{ran}(W_D^\dagger)$. Applying the finite linear map $\bigwedge^4 C_R$ by congruence preserves positivity exactly. Applying the positive lift $\mathcal{L}_4$ and integrating then yields

$$K_4 \le T_4 + \varepsilon_{\mathrm{F}} R_4, \qquad R_4 = \int_T \sum_{D=1}^{23} \sum_{r\in\mathcal{R}_D} A(w^T_{D,r}(R))^\dagger A(w^T_{D,r}(R))\ \mathrm{d}\mu_2(R). \tag{5.34}$$

In particular, the proof never inverts $G_D$ and never perturbs its zero eigenvalues. It transports a positive physical four-body operator, rather than asserting positivity of $A_D$ on the entire copy space.

The allowance $R_4$ also has a particle-number-independent pair-energy bound. For fixed $D$, the coefficient rows $S_r(D, D; p, j, k)$ are orthonormal in the formal input coordinates $(p, j, k)$ with $p + j + k = D$ and $j < k$. This follows by applying the two orthogonal changes of variables in (5.18) in succession.

Bessel's inequality remains valid when those coordinates are arbitrary Hilbert-space vectors, so

$$\sum_{r\in\mathcal{R}_D}\|A(w_{D,r}^T(R))\psi\|^2 \le \sum_{\substack{p+j+k=D\\ j<k}}\|c_k(R)c_j(R)B_p(R)\psi\|^2$$
$$\le (1+e_{24})^2\sum_{\substack{p+j+k=D\\ j<k}}\|B_p(R)\psi\|^2. \tag{5.35}$$

The last step uses the one-particle CAR norm bound twice. It does not assume that physical torus jets or their wedge images are orthogonal. For $n=j+k\ge 1$, there are exactly $\lceil n/2\rceil$ nonnegative pairs $j<k$ with that sum, and $p=D-n$ is determined. Hence

$$n_D=\sum_{n=1}^{D}\left\lceil\frac{n}{2}\right\rceil=\left\lfloor\frac{(D+1)^2}{4}\right\rfloor,\qquad \sum_{D=1}^{23}n_D=1222. \tag{5.36}$$

Integrating every pair term in (5.35) with the exact pair-frame identity proves

$$R_4\le 1222(1+e_{24}(\Lambda))^2H,\qquad K_4\le T_4+1222\varepsilon_{\mathrm{F}}(1+e_{24}(\Lambda))^2H. \tag{5.37}$$

## 5.6 Closing the fermionic gap estimate

We collect the estimates as quadratic forms. The exact normal-ordering identity is $H^2=\mathcal{L}_2(h_T^2)+C_3+S_4$. The all-COM contact-frame estimate gives $\mathcal{L}_2(h_T^2)\ge(1-\varepsilon_2^{\mathrm{F}}(\Lambda))H$; it does not replace $h_T^2$ by $h_T$. Combining this bound with (5.28) and (5.8) yields

$$H^2\ge K_3^{(0)}+T_4+[1-\varepsilon_2^{\mathrm{F}}-\delta_\Lambda-C_4^{\mathrm{F}}\varepsilon_4^{\mathrm{F}}]H. \tag{5.38}$$

Next, (5.26), (5.30), and (5.37) imply

$$K_3^{(0)}+T_4\ge K_T-[\eta+C_\alpha^{\mathrm{F}}e_8(1+e_8)+1222\varepsilon_{\mathrm{F}}(1+e_{24})^2]H. \tag{5.39}$$

Consequently

$$H^2\ge K_T+\Gamma_{\mathrm{F}}(\Lambda)H\ge\Gamma_{\mathrm{F}}(\Lambda)H, \tag{5.40}$$
$$\Gamma_{\mathrm{F}}(\Lambda)=1-\eta-\delta_\Lambda-1222\varepsilon_{\mathrm{F}}(1+e_{24}(\Lambda))^2-\varepsilon_2^{\mathrm{F}}(\Lambda)$$
$$-C_\alpha^{\mathrm{F}}e_8(\Lambda)(1+e_8(\Lambda))-C_4^{\mathrm{F}}(\Lambda)\varepsilon_4^{\mathrm{F}}(\Lambda). \tag{5.41}$$

Every term in this scalar budget has been paid for by the same operator $H$. Fixed-degree Gram errors and the two finite relative-frame errors tend to zero uniformly as the actual shortest nonzero lattice vector tends to infinity. Equation (5.13) gives the remaining limit. Therefore

$$\lim_{\ell(\Lambda)\to\infty}\Gamma_{\mathrm{F}}(\Lambda)=1-\frac{93527408868499}{10^{14}}-\frac{61}{4096}-1222\frac{3}{10^6}=\frac{4616733319001}{10^{14}}=:\gamma_{\mathrm{F}}^*. \tag{5.42}$$

Given $0<\gamma<\gamma_{\mathrm{F}}^*$, uniform convergence supplies a systole threshold $L_\gamma$ for which $\Gamma_{\mathrm{F}}(\Lambda)\ge\gamma$. Then (5.40) proves $H^2\ge\gamma H$ on every particle sector and hence on the full fermionic Fock space. A positive eigenvalue $E$ satisfies $E^2\ge\gamma E$, so $E\ge\gamma$. This proves the gap assertion of Theorem 2.2. The argument supplies an asymptotic threshold, not an explicit numerical value of $L_\gamma$.

In particular, $1/25<\gamma_{\mathrm{F}}^*$. At $M=3N$, the common zero-mode argument identifies the three-dimensional Laughlin ground space, and the same inequality gives a gap at least $1/25$ once the systole is sufficiently large. Particle number, magnetic flux, and twist do not enter the error constants. The geometric condition remains essential: neither increasing area alone nor choosing two long but nearly parallel lattice generators ensures a large shortest cycle.

# 6 The bosonic bound: positive comparison and a finite certificate

We now prove the bosonic assertion of Theorem 2.1. Throughout this section, $H = H_B$ denotes the full contact interaction, with planar nonzero two-particle eigenvalue one. We also abbreviate the bosonic connected and zero-contraction terms by $S_3$ and $S_4$, and write $H_k^{\mathrm{pl}} = H_{k,B}^{\mathrm{pl}}$. The exact identities established in the preceding sections reduce the problem to controlling the negative three-body tail and the errors of a finite sum-of-squares certificate. Bosonic occupation numbers are unbounded, so those errors must be compared with the positive operators $H$ and $S_4$, rather than with a particle-number-dependent multiple of the identity. We obtain that comparison from coherent-state occupation inequalities and positive Gaussian kernels.

All operator inequalities in the proof are first understood on a fixed particle sector. The physical torus lowest-Landau-level space is finite dimensional, and every such sector is finite dimensional. The passage to the full bosonic Fock space is made at the end of the section. The periodized orbitals remain the raw jets of Section 3; none is orthogonalized or pointwise renormalized in the definitions of the interaction or the certificate.

## 6.1 Coherent occupation inequalities

The coherent-occupation comparison is adapted from the bosonic spherical candidate of Shen [15]. The constants $k_1, k_2, c, s$ below are torus reproducing-kernel quantities. Their bounds and the positive thermal transfer must therefore be established for this geometry, independently of that source's finite-sphere interval verification.

Let $K_j^T$ be the charge-$j$ parameter reproducing kernel, and set

$$k_j(R) = K_j^T(R,R), \qquad s = \sup_{R\in T} k_1(R), \qquad c = \sup_{R\in T} \frac{k_2(R)}{k_1(R)^2}. \tag{6.1}$$

In the large-systole regime used below, $k_1$ is strictly positive. Write $B_R = B_0^B(R)$ and $\beta_R = u_0^T(R)/\sqrt{k_1(R)}$. The latter orbital is normalized only for the purpose of applying the following occupation estimates.

**Lemma 6.1** (One and two coherent spectators)**.** *For a normalized coherent orbital $\beta = \beta_R$, let $n_\beta = a(\beta)^\dagger a(\beta)$. Then*

$$\frac{1}{2} n_\beta(n_\beta - 1) \le cH, \qquad n_\beta \le cH + I. \tag{6.2}$$

*For any two such orbitals $\beta, \gamma$, including nonorthogonal ones,*

$$\big(a(\gamma)a(\beta)\big)^\dagger a(\gamma)a(\beta) \le 4cH + 2I. \tag{6.3}$$

*Proof.* The two-particle evaluation map $Q$ has $h_T = Q^\dagger Q$, and $Q\psi$ is a charge-two parameter lowest-Landau-level section. Its reproducing property gives $|Q\psi(R)|^2 \le k_2(R)\|Q\psi\|^2$. Consequently, $|v_0(R)\rangle\langle v_0(R)| \le k_2(R) h_T$. Applying the positive lift $\mathcal{L}_2$ yields $B_R^\dagger B_R \le k_2(R) H$. Since

$$B_R = \frac{k_1(R)}{\sqrt{2}}\, a(\beta_R)^2, \qquad B_R^\dagger B_R = \frac{k_1(R)^2}{2} n_{\beta_R}(n_{\beta_R} - 1), \tag{6.4}$$

the first inequality in (6.2) follows. The scalar inequality $n \le n(n-1)/2 + 1$ for every nonnegative integer $n$ gives the second inequality by the spectral calculus for $n_\beta$. In particular, the same statement applies in the output sector after $B_R$ has removed two particles.

For the two-orbital estimate, put $A = a(\gamma)a(\beta)$ and $\omega = \langle \beta, \gamma \rangle$, so $|\omega| \le 1$. The canonical commutation relations give

$$\begin{aligned} A^\dagger A &= \frac{1}{2}(n_\beta n_\gamma + n_\gamma n_\beta) - \mathrm{Re}\big(\omega a(\beta)^\dagger a(\gamma)\big) \\ &\le \frac{1}{2}(n_\beta^2 + n_\gamma^2) + \frac{1}{2}(n_\beta + n_\gamma) \le 4cH + 2I. \end{aligned} \tag{6.5}$$

The first inequality uses $(n_\beta - n_\gamma)^2 \ge 0$ and quadratic-form Cauchy–Schwarz on the cross term. For the second, apply $n_x^2 \le 2cH + n_x$ and then $n_x \le cH + I$ for $x = \beta, \gamma$. This proves the lemma without a bound on an annihilation operator's norm. □

Replacing a normalized spectator by the raw coherent jet introduces at most a factor $s$; replacing two spectators introduces at most $s^2$. No additional factor is attached to the pair anchor. Indeed, its subsequent integration is governed by the exact identities

$$\int_T B_\alpha^\dagger B_\alpha \, \mathrm{d}\mu_2(\alpha) = H, \qquad \int_T B_\alpha^\dagger H B_\alpha \, \mathrm{d}\mu_2(\alpha) = S_4. \tag{6.6}$$

The second identity follows by substituting the coherent integral for $H$ and commuting the pair annihilators. Thus the occupation estimates pay for an arbitrary spectator population using precisely the positive zero-contraction term already present in $H^2$.

## 6.2 Positive thermal kernels and their exact torus transfer

The planar coherent-comparison input is adapted from Shen [15]. The exact transfer of positive thermal kernels to the magnetic quotient is established below using the convergent domain of Section 3; it is not a consequence of the source's spherical Schur averages.

The auxiliary spaces in this subsection are planar spaces. Let $U$ be the planar one-particle Bargmann space and let $V$ be the pair subspace with relative degree zero, with orthonormal center-of-mass basis $v_p$. A normalized coherent vector $|\sqrt{2}\alpha\rangle_V$ represents the pair vector $B_\alpha^\dagger|0\rangle$ on the plane. The insertion maps are

$$\begin{aligned} &W_3 : V \otimes U \longrightarrow \mathrm{Sym}^3 U, && W_3(v_p \otimes u) = B_p^\dagger u, \\ &W_4 : V \otimes \mathrm{Sym}^2 U \longrightarrow \mathrm{Sym}^4 U, && W_4(v_p \otimes u) = B_p^\dagger u. \end{aligned} \tag{6.7}$$

These maps use the normalized symmetric-tensor to Fock-space identification. In particular, $W_3 = \sqrt{3} P_{\mathrm{sym}} \iota_3$, where $\iota_3$ is the isometric embedding of the distinguished pair and the spectator before symmetrization. Moreover,

$$W_k W_k^\dagger = H_k^{\mathrm{pl}}, \qquad \|W_3\| \le \sqrt{3}, \qquad \|W_4\| \le \sqrt{6}. \tag{6.8}$$

The first identity is the sum over all physical pairs. The norm bounds follow because $H_k^{\mathrm{pl}}$ is the sum of $\binom{k}{2}$ pair projections. Each $W_k$ intertwines the total lowering and raising operators, since inserting a pair is compatible with simultaneous magnetic translation. Hence in the planar center-of-mass decomposition it has the form $I_{\mathrm{COM}} \otimes W_{k,\mathrm{rel}}$, including the full relative multiplicity space.

For clarity, the center-of-mass factors used in the transfer are recorded explicitly. The planar twirl of a trace-class anchored seed $\sigma$ and its torus counterpart are

$$\int_{\mathbb{C}} T_\alpha^{\otimes k} \sigma T_\alpha^{\otimes k\dagger} \, \mathrm{d}\mu_2(\alpha) = \frac{2}{k} I_{\mathrm{COM}} \otimes \mathrm{Tr}_{\mathrm{COM}} \sigma, \tag{6.9}$$

$$\int_T C_\alpha^{(k)} \sigma C_\alpha^{(k)\dagger} \, \mathrm{d}\mu_2(\alpha) = \Omega_k\Big(\frac{2}{k} \mathrm{Tr}_{\mathrm{COM}} \sigma\Big). \tag{6.10}$$

For finite-degree seeds, the second identity is the integrated ladder identity of Lemma 3.1. The factor $2/k$ in the first identity comes from the normalized center-of-mass displacement $h = \sqrt{k}\alpha$, so that $\mathrm{d}\mu_2(\alpha) = (2/k)\,\mathrm{d}^2h/\pi$. The Weyl average traces out the center-of-mass oscillator and leaves the relative operator unchanged, rather than replacing it by a scalar. We justify the extension of (6.10) to the thermal seeds below. No global relative-coordinate decomposition on the torus is assumed.

### 6.2.1 One spectator

For $a > 0$, define a positive one-particle seed by

$$R_a = \int_{\mathbb{C}} e^{-a|\beta|^2} |u_0(\beta)\rangle\langle u_0(\beta)|\ \mathrm{d}\mu_1(\beta) = \sum_{j=0}^{\infty} (1+a)^{-j-1} |e_j\rangle\langle e_j|. \tag{6.11}$$

The equality follows by expanding the normalized coherent vector in the basis $e_j$: the angular integral makes the matrix diagonal, and the radial Gaussian moment gives $(1+a)^{-j-1}$. In particular, $R_a$ is positive and trace class. Set

$$\sigma_3 = |0\rangle_V\langle 0| \otimes R_a, \qquad \tilde{\sigma}_3 = W_3 \sigma_3 W_3^{\dagger}. \tag{6.12}$$

In the planar simultaneous-translation integral, introduce the coherent coordinates $u = \sqrt{2}\alpha$ and $v = \alpha + \beta$, followed by the unitary Jacobi change of variables

$$\zeta = \frac{\sqrt{2}u + v}{\sqrt{3}}, \qquad \xi = \frac{-u + \sqrt{2}v}{\sqrt{3}}. \tag{6.13}$$

The measure is $\mathrm{d}\mu_2(\alpha)\,\mathrm{d}\mu_1(v) = \mathrm{d}^2u\,\mathrm{d}^2v/\pi^2$, and the Gaussian weight becomes $\exp[-(3a/2)|\xi|^2]$. The $\zeta$ integral resolves the center-of-mass identity, while (6.11) evaluates the $\xi$ integral. Thus, with $r = 2/(2+3a)$, the auxiliary twirl is exactly $I_{\mathrm{COM}} \otimes r^{n_\xi+1}$. Intertwining with $W_3$ and comparing with (6.9) gives

$$\frac{2}{3}\,\mathrm{Tr}_{\mathrm{COM}}\,\tilde{\sigma}_3 = W_{3,\mathrm{rel}}\, r^{n_\xi+1} W_{3,\mathrm{rel}}^{\dagger}. \tag{6.14}$$

Apply $\Omega_3$ to this planar relative coefficient and use (6.10). Substituting the positive coherent integral for $R_a$ then gives an explicit physical coefficient. Indeed, simultaneous translation by $\alpha$ sends the spectator coherent vector at offset $\beta$ to the periodized vector at $\alpha + \beta$, multiplied by $\exp[-i\,\mathrm{Im}(\alpha\overline{\beta})]$. This phase cancels in the rank-one projector. Positive lifting therefore gives the exact identity

$$\begin{aligned}\mathcal{G}_3(a) &:= \mathcal{L}_3\Omega_3\left(W_{3,\mathrm{rel}}\, r^{n_\xi+1} W_{3,\mathrm{rel}}^{\dagger}\right)\\ &= \int_{\alpha\in T} \mathrm{d}\mu_2(\alpha) \int_{\beta\in\mathbb{C}} \mathrm{d}\mu_1(\beta)\, e^{-a|\beta|^2} B_\alpha^{\dagger} a\big(u_0^T(\alpha+\beta)\big)^{\dagger} a\big(u_0^T(\alpha+\beta)\big) B_\alpha.\end{aligned} \tag{6.15}$$

The spectator offset ranges over the whole plane. Restricting that integral to one torus cell without retaining its images would change the coefficient and is not part of the transfer. Lemma 6.1 bounds the spectator factor by $s(cH + I)$, and $\int_{\mathbb{C}} e^{-a|\beta|^2}\,\mathrm{d}\mu_1(\beta) = 1/a$. The exact anchor identities (6.6) imply

$$\mathcal{G}_3(a) \le \frac{s}{a}(cS_4 + H). \tag{6.16}$$

### 6.2.2 Two spectators

On $V \otimes \mathrm{Sym}^2 U$, use the seed

$$\sigma_4 = |0\rangle_V\langle 0| \otimes [R_a \otimes R_a]_{\mathrm{sym}}, \qquad \tilde{\sigma}_4 = W_4 \sigma_4 W_4^{\dagger}, \tag{6.17}$$

where $[\cdot]_{\mathrm{sym}}$ denotes compression to the normalized symmetric tensor space. Set $u = \sqrt{2}\alpha$, $v = \alpha + \beta$, and $w = \alpha + \gamma$. The unitary Jacobi coordinates are

$$\zeta = \frac{\sqrt{2}u + v + w}{2}, \qquad \xi_1 = \frac{v - w}{\sqrt{2}}, \qquad \xi_2 = \frac{u}{\sqrt{2}} - \frac{v + w}{2}. \tag{6.18}$$

The measure is preserved, and the weight separates because

$$|\beta|^2 + |\gamma|^2 = |\xi_1|^2 + 2|\xi_2|^2. \tag{6.19}$$

The center-of-mass coherent integral gives the identity; the two remaining Gaussian moments give $r_1^{n_1+1} r_2^{n_2+1}$ with $r_1 = (1+a)^{-1}$ and $r_2 = (1+2a)^{-1}$. Spectator interchange sends $\xi_1$ to $-\xi_1$ and fixes $\xi_2$, so the symmetric spectator space is the subspace with even $n_1$. Equation (6.9), now with $k = 4$, therefore yields

$$\frac{1}{2}\,\mathrm{Tr}_{\mathrm{COM}}\,\tilde{\sigma}_4 = W_{4,\mathrm{rel}}\left[r_1^{n_1+1} r_2^{n_2+1}\right] W_{4,\mathrm{rel}}^{\dagger}. \tag{6.20}$$

Transfer this relative coefficient by $\Omega_4$ and then lift it. The resulting positive integral is

$$\begin{aligned}
\mathcal{G}_4(a) &:= \mathcal{L}_4\Omega_4\left(W_{4,\mathrm{rel}}[r_1^{n_1+1} r_2^{n_2+1}]W_{4,\mathrm{rel}}^{\dagger}\right) \\
&= \frac{1}{2}\int_{\alpha\in T} \mathrm{d}\mu_2(\alpha) \int_{\beta,\gamma\in\mathbb{C}} \mathrm{d}\mu_1(\beta)\,\mathrm{d}\mu_1(\gamma)\, e^{-a(|\beta|^2+|\gamma|^2)} B_\alpha^{\dagger} A_{\alpha;\beta\gamma}^{\dagger} A_{\alpha;\beta\gamma} B_\alpha, \\
A_{\alpha;\beta\gamma} &= a(u_0^T(\alpha+\gamma))a(u_0^T(\alpha+\beta)).
\end{aligned} \tag{6.21}$$

The factor $1/2$ in this integral has a different origin from the factor $1/2$ in (6.20). The latter is the center-of-mass factor $2/k$. The former comes from the symmetric-tensor identification

$$P_{\mathrm{sym}}^{(2)}(|\beta\rangle \otimes |\gamma\rangle) \longleftrightarrow \frac{a(\beta)^{\dagger} a(\gamma)^{\dagger}|0\rangle}{\sqrt{2}}. \tag{6.22}$$

Both factors have therefore been included, once each. The two offsets in (6.21) again range over the entire plane. Applying (6.3) to the unnormalized jets costs at most $s^2(4cH + 2I)$; the two Gaussian marginals give $1/a^2$. Together with (6.6), this proves

$$\mathcal{G}_4(a) \le \frac{s^2}{a^2}(2cS_4 + H). \tag{6.23}$$

### 6.2.3 Convergence and preservation of positive order

The preceding formulas can be obtained without treating infinite periodization as a bounded operator on the entire planar Bargmann space. To see this, truncate each anchored seed by total degree $d$. Before insertion, the degree-$d$ traces obey

$$\mathrm{Tr}(\sigma_{3,d}) \le (1+a)^{-d-1}, \qquad \mathrm{Tr}(\sigma_{4,d}) \le (d+1)(1+a)^{-d-2}. \tag{6.24}$$

Insertion multiplies these bounds by at most 3 and 6, respectively, by (6.8). The raw-jet estimate of Lemma 3.2 bounds the squared norm of $C_R^{(k)}$ on a degree-$d$ block by $C_k(d+1)^{2k}$, uniformly in $R$ on the fixed torus. Consequently,

$$\sum_{d=0}^{\infty} \sup_{R\in T} \mathrm{Tr}\big(C_R^{(k)} \tilde{\sigma}_{k,d} C_R^{(k)\dagger}\big) < \infty, \qquad k = 3, 4. \tag{6.25}$$

The fundamental cell has finite area. Thus finite-degree identities pass to the limit in trace norm on the physical few-body space, and the center-of-mass partial trace and coherent integrals may be interchanged

with that limit. On the coherent side, boundedness of $\sup_R \|u_0^T(R)\|$ and Gaussian integrability also give direct domination. This proves (6.15) and (6.21) with all offsets retained.

Every relative comparison below is diagonal in total relative degree, although its blocks may have nontrivial multiplicity. On each finite collection of blocks, take the positive difference of the two coefficients and apply the congruence defining $\Omega_k$. The resulting operator remains positive. The exponential domination just established then passes this inequality through a positive limit. Hence the thermal comparisons preserve order on the torus without defining $\Omega_k$ on arbitrary nondecaying infinite relative operators.

### 6.3 The negative tail and two finite comparison operators

Let $\xi_z = \xi_z^B$ be the normalized pair–spectator relative oscillator vector of degree $z$, as in the three-body calculation leading to (4.18). Put

$$T_9 = \mathcal{L}_3\Omega_3\left(W_{3,\mathrm{rel}}\left[2\sum_{\substack{z\ge 9\\ z\ \mathrm{odd}}} 2^{-z}|\xi_z\rangle\langle\xi_z|\right]W_{3,\mathrm{rel}}^{\dagger}\right), \tag{6.26}$$

$$R_{k,D} = \mathcal{L}_k\Omega_k\left(H_{k,\mathrm{rel}}^{\mathrm{pl}}\Pi_{\le D}^{(k)}\right), \qquad (k,D) = (3,8),(4,16). \tag{6.27}$$

The projections in (6.27) retain every physical relative multiplicity at the indicated degree. They are not projections onto only one selected vector in each degree.

**Proposition 6.2** (Bosonic comparison constants)**.** *The operators just defined satisfy*

$$T_9 \le c_9 s(cS_4 + H), \qquad c_9 = 30\left(\frac{5}{9}\right)^9 = \frac{19531250}{129140163}, \tag{6.28}$$

$$R_{3,8} \le C_3^B s(cS_4 + H), \qquad C_3^B = 12\left(\frac{9}{8}\right)^9 = \frac{1162261467}{33554432}, \tag{6.29}$$

$$R_{4,16} \le C_4^B s^2(2cS_4 + H), \qquad C_4^B = 272\left(\frac{9}{8}\right)^{17} = \frac{283512088894331673}{140737488355328}. \tag{6.30}$$

*Proof.* For $r = 9/10$, comparison of each relative eigenvalue gives

$$2\sum_{\substack{z\ge 9\\ z\ \mathrm{odd}}} 2^{-z}|\xi_z\rangle\langle\xi_z| \le \frac{2}{r}\left(\frac{1}{2r}\right)^9 r^{n_\xi+1}. \tag{6.31}$$

At $z = 9$ the corresponding scalar coefficients agree; for higher odd $z$ their ratio decreases because $2r > 1$, and the left-hand coefficient is zero in every other degree. Choosing $a = 2/27$ gives $r = 2/(2+3a)$. Apply the positive maps $W_{3,\mathrm{rel}}(\cdot)W_{3,\mathrm{rel}}^{\dagger}$, $\Omega_3$, and $\mathcal{L}_3$, in that order. Equation (6.16) then proves (6.28), because $(2/(ar))(1/(2r))^9 = 30(5/9)^9$.

For the finite three-body operator, take $a = 1/12$ and $r = 8/9$. On relative degrees $0 \le z \le 8$, $r^{z+1} \ge r^9$, so $\Pi_{\le 8}^{\mathrm{aux}} \le r^{-9}r^{n_\xi+1}$. Since $W_3$ intertwines the center-of-mass oscillator and preserves total degree, (6.8) also gives

$$W_{k,\mathrm{rel}}\Pi_{\le D}^{\mathrm{aux}}W_{k,\mathrm{rel}}^{\dagger} = H_{k,\mathrm{rel}}^{\mathrm{pl}}\Pi_{\le D}^{(k)}, \qquad k = 3,4. \tag{6.32}$$

Applying the same positive maps and (6.16) gives (6.29), with $r^{-9}/a = 12(9/8)^9$.

For four particles, use the full auxiliary space $V \otimes \mathrm{Sym}^2 U$ from (6.7). On $n_1 + n_2 \le 16$ with $n_1$ even, the smallest eigenvalue of $r_1^{n_1+1}r_2^{n_2+1}$ is $r_1r_2^{17}$: $r_2 \le r_1 < 1$, so the minimum places all sixteen relative quanta in the second oscillator. Hence the auxiliary cutoff is bounded by $(r_1r_2^{17})^{-1}$ times the thermal coefficient. Apply (6.32) and (6.23), and set $a = 1/16$. Then $r_1 = 16/17$, $r_2 = 8/9$, and

$$\frac{1}{a^2 r_1 r_2^{17}} = 272\left(\frac{9}{8}\right)^{17}, \tag{6.33}$$

which proves (6.30). All comparisons were established at the positive few-body coefficient level before lifting, so their constants do not depend on the number of particles. □

### 6.4 The fixed 138-row certificate

The finite input is the planar row certificate supplied with the bosonic candidate manuscript of Shen [15]. Only its fixed rational row data and the finite planar block inequalities are used here. No spherical gap theorem or finite-flux interval estimate is an assumption. Appendix B gives the exact data decoding, occupation metric, relative-block construction, and rational positivity test.

For each row $\rho$, the channel $t_\rho$ belongs to $\{0,1,2,3\}$. The respective row counts are $31, 37, 36, 34$. Define, using raw periodized jets,

$$F^\rho(R) = \lambda_\rho B^B_{t_\rho}(R) + \sum_{\substack{p,j\ge 0\\ t_\rho\le p+j\le 8}} \alpha^\rho_{pj} a_{p+j-t_\rho}(R)^\dagger a_j(R) B^B_p(R), \tag{6.34}$$

$$K = \int_T \sum_\rho F^\rho(R)^\dagger F^\rho(R)\ \mathrm{d}\mu_2(R) \ge 0. \tag{6.35}$$

The frozen integers divided by $10^{12}$ specify the rational numbers $y^\rho_0$ and $y^\rho_{pj}$. Their relation to the physical coefficients is

$$y^\rho_0 = \lambda_\rho \kappa_{t_\rho}, \qquad y^\rho_{pj} = \alpha^\rho_{pj}\kappa_p\sqrt{\frac{j!}{i!}}, \quad i = p+j-t_\rho, \qquad \kappa_p^2 = \frac{p!}{2^{p+1}}, \quad \kappa_p > 0. \tag{6.36}$$

Thus all real algebraic row coefficients are uniquely fixed. On the plane, the canonical polynomial variables $x_j = \sqrt{j!}\,a_j^\dagger$ and $\partial_j = a_j/\sqrt{j!}$ satisfy $[\partial_i, x_j] = \delta_{ij}$. In these variables $B_p = \kappa_p \widehat{B}_p$, where $\widehat{B}_p = \sum_{a+b=p}\partial_a\partial_b$, and each row becomes the rational differential operator

$$F^\rho = y^\rho_0 \widehat{B}_{t_\rho} + \sum_{p,j} y^\rho_{pj} x_{p+j-t_\rho}\partial_j \widehat{B}_p. \tag{6.37}$$

Physical adjoints in these coordinates are taken with the occupation metric $\prod_j n_j!(j!)^{n_j}$, not with the Euclidean metric of the coefficient vector.

Suppress the row label temporarily and put $i = p+j-t$, $i' = p'+j'-t$. With canonical planar commutators, the normal-ordered contributions beyond $\lambda^2 B_t^\dagger B_t$ are exactly

$$\begin{aligned}\mathsf{K}_3^{\mathrm{row}} &= \lambda\sum_{p,j}\alpha_{pj}\left(B_t^\dagger a_i^\dagger a_j B_p + (B_t^\dagger a_i^\dagger a_j B_p)^\dagger\right)\\ &\quad + \sum_{p,j,p',j'}\alpha_{pj}\alpha_{p'j'}\delta_{ii'}B_p^\dagger a_j^\dagger a_{j'}B_{p'},\end{aligned} \tag{6.38}$$

$$\mathsf{K}_4^{\mathrm{row}} = \sum_{p,j,p',j'}\alpha_{pj}\alpha_{p'j'}B_p^\dagger a_j^\dagger a_{i'}^\dagger a_i a_{j'}B_{p'}. \tag{6.39}$$

The first coefficient has total orbital degree at most eight; the second has total orbital degree at most sixteen. Let $\overline{\mathsf{K}}_{k,\mathrm{rel}}$ be their summed center-of-mass partial traces with the factor $2/k$ from (6.10). The exact finite certificate is

$$S^{\mathrm{pl}}_{3,\mathrm{rel}}\Pi^{(3)}_{\le 8} - \overline{\mathsf{K}}_{3,\mathrm{rel}} + \delta_B H^{\mathrm{pl}}_{3,\mathrm{rel}}\Pi^{(3)}_{\le 8} \ge 0, \tag{6.40}$$

$$\alpha_B S^{\mathrm{pl}}_{4,\mathrm{rel}}\Pi^{(4)}_{\le 16} - \overline{\mathsf{K}}_{4,\mathrm{rel}} + \delta_B H^{\mathrm{pl}}_{4,\mathrm{rel}}\Pi^{(4)}_{\le 16} \ge 0, \tag{6.41}$$

$$\delta_B = 10^{-6}, \qquad \mu_B = \frac{1}{50}, \qquad \alpha_B = 1 - c_9 - \mu_B.$$

These are $9+17 = 26$ rational block tests, including the zero-dimensional degree-one blocks. Rational Schur elimination verifies positive semidefiniteness; in each block the exact kernel of $H^{\mathrm{pl}}_k$ is preserved, and the positive rank equals that of $H^{\mathrm{pl}}_k$. The equations define finite-dimensional statements with no limiting tolerance. Their construction and the block dimensions and ranks are given in Appendix B.

The two-body row cost is

$$\eta = \sum_{\rho} \lambda_{\rho}^2 = \frac{76419170375082343297845 7}{150000000000000000000000000}. \tag{6.42}$$

Apply the positive periodization map and then the positive lift to (6.40)–(6.41). Define

$$K_3^0 = \mathcal{L}_3\Omega_3(\overline{\mathsf{K}}_{3,\mathrm{rel}}), \qquad K_4 = \mathcal{L}_4\Omega_4(\overline{\mathsf{K}}_{4,\mathrm{rel}}), \qquad T_4 = \mathcal{L}_4\Omega_4(S_{4,\mathrm{rel}}^{\mathrm{pl}}\Pi_{\le 16}^{(4)}). \tag{6.43}$$

This transfer is a congruence of coefficient tensors, followed by integration. It therefore requires neither an inverse torus few-body Gram matrix nor a choice of basis for its varying kernel. The exact connected formula (4.18) has nonnegative even-degree tail and negative odd-degree tail $-T_9$. Consequently, the transferred inequalities imply

$$S_3 \ge K_3^0 - \delta_B R_{3,8} - T_9, \qquad K_4 \le \alpha_B T_4 + \delta_B R_{4,16}. \tag{6.44}$$

## 6.5 The raw-jet commutator error in Loewner order

On the torus, $[a_i(R), a_{i'}(R)^\dagger] = g_{ii'}(R)$. In the normal ordering of a row square, the only change from (6.38)–(6.39) is the replacement of the single contraction $\delta_{ii'}$ by $g_{ii'}(R)$. The zero-contraction four-body coefficient is unchanged. The exact pair ladder identity integrates the two-body coefficient to $\eta H$, so

$$K = \eta H + K_3^0 + K_4 + R_g. \tag{6.45}$$

Let

$$\begin{aligned} e_8 &= \sup_{R\in T} \|g_{0,\dots,8}(R) - I\|, \\ C_{\mathrm{rows}}^B &= \sum_{\rho,p,j} (\alpha_{pj}^{\rho})^2 = \frac{3117157632914067841282955 7}{875000000000000000000000}, \qquad \kappa_B = 6C_{\mathrm{rows}}^B e_8. \end{aligned} \tag{6.46}$$

**Lemma 6.3** (Two-sided commutator estimate)**.**

$$-\kappa_B R_{3,8} \le R_g \le \kappa_B R_{3,8}. \tag{6.47}$$

*Proof.* At a fixed center, write $X_{pj} = a_j B_p^B$ and, for each row, collect $Y_i^{\rho} = \sum_{p+j-t_\rho=i} \alpha_{pj}^{\rho} X_{pj}$. The local remainder is the Hermitian quadratic form

$$R_g(R) = \sum_{\rho,i,i'} (Y_i^{\rho})^\dagger (g_{ii'}(R) - \delta_{ii'}) Y_{i'}^{\rho}. \tag{6.48}$$

The matrix inequality $-e_8 I \le g - I \le e_8 I$ is valid for all indices occurring here. Apply it to the vector of output states $Y_i^{\rho}\psi$ and then apply Cauchy–Schwarz to each finite coefficient sum. This gives the pointwise Loewner bounds

$$-\mathcal{E}(R) \le R_g(R) \le \mathcal{E}(R), \qquad \mathcal{E}(R) = e_8 C_{\mathrm{rows}}^B \sum_{p+j\le 8} (a_j B_p^B)^\dagger (a_j B_p^B). \tag{6.49}$$

There is no estimate of $\|a_j\|$ in this step. To integrate the envelope, use the auxiliary orthonormal vectors $v_p \otimes e_j$. Its coefficient is $W_3 P_{\mathrm{degree}\le 8}^{\mathrm{aux}} W_3^\dagger$. A relative oscillator state of degree $z \le 8$ has exactly $9 - z$ center-of-mass descendants in this cutoff. Equation (6.10) therefore assigns it the partial-trace coefficient $(2/3)(9 - z) \le 6$. Intertwining with $W_3$, preserving positive order, and lifting now give

$$\int_T \mathcal{E}(R)\ \mathrm{d}\mu_2(R) \le 6 e_8 C_{\mathrm{rows}}^B R_{3,8}. \tag{6.50}$$

Integrating both inequalities in (6.49) proves (6.47). Thus the raw-jet metric error remains a three-body error with a particle-number-independent coefficient. □

### 6.6 Assembly of the bosonic inequality

Use the exact decomposition $H^2 = \mathcal{L}_2(h_T^2) + S_3 + S_4$ from Proposition 4.1. The pair almost-projector estimate gives $\mathcal{L}_2(h_T^2) \geq (1-\epsilon_2)H$. The relative frame bound of Lemma 3.3, with charges two and two and relative degrees up to sixteen, gives $T_4 \leq (1+\epsilon_4)S_4$. In terms of the raw pair descendants, that finite four-body coefficient is explicitly

$$C_d(R) = \sum_{p=0}^{d} (-1)^{d-p} \sqrt{\binom{d}{p}}\, 2^{-d/2} B_p^B(R) B_{d-p}^B(R),$$
$$T_4 = \int_T \sum_{d=0}^{16} C_d(R)^\dagger C_d(R)\ \mathrm{d}\mu_4(R) \leq (1+\epsilon_4)S_4. \tag{6.51}$$

Odd $d$ vanish because the pair annihilators commute. The estimate retains the error as a multiple of $S_4$, which is the positive term available in the bosonic argument.

Substitute (6.44) into the connected decomposition, then eliminate $K_3^0 + K_4$ using (6.45). Since $\alpha_B > 0$, the four-body frame comparison is used in the correct direction. Finally use the upper bound on $R_g$ from Lemma 6.3. The result is

$$\begin{aligned} H^2 \geq K &+ (1-\epsilon_2-\eta)H + [1-\alpha_B(1+\epsilon_4)]S_4 \\ &- (\delta_B+\kappa_B)R_{3,8} - \delta_B R_{4,16} - T_9. \end{aligned} \tag{6.52}$$

Apply Proposition 6.2 to the three negative terms. This gives

$$H^2 \geq K + A_\Lambda H + B_\Lambda S_4, \tag{6.53}$$

$$A_\Lambda = 1 - \epsilon_2 - \eta - s\big[c_9 + (\delta_B+\kappa_B)C_3^B\big] - \delta_B C_4^B s^2, \tag{6.54}$$
$$B_\Lambda = 1 - \alpha_B(1+\epsilon_4) - sc\big[c_9 + (\delta_B+\kappa_B)C_3^B\big] - 2\delta_B C_4^B s^2 c. \tag{6.55}$$

As the actual shortest nonzero lattice vector tends to infinity, the common frame estimates give $\epsilon_2, \epsilon_4, e_8 \to 0$ and $s, c \to 1$, uniformly in twists and independently of the displayed lattice basis. Therefore,

$$\begin{aligned} A_\infty &= 1 - \eta - c_9 - \delta_B(C_3^B + C_4^B) \\ &= \frac{233820146330244154759885264985357404555 97}{693315970856386560000000000000000000000000} \approx 0.3372490410705938 4 > \frac{1}{3}, \end{aligned} \tag{6.56}$$
$$B_\infty = \mu_B - \delta_B(C_3^B + 2C_4^B) = \frac{1121425355698906343}{70368744177664000000} \approx 0.015936412803780887 > 0. \tag{6.57}$$

Since $K$ and $S_4$ are positive, every $\gamma < A_\infty$ admits a systole threshold for $H^2 \geq \gamma H$. We next give a fully explicit sufficient threshold for $\gamma = 1/3$.

### 6.7 An explicit rational budget at systole twenty

Assume $\ell(\Lambda) \geq L = 20$ in the dimensionless coordinate $z = x/(\sqrt{2}\ell_B)$. Partition the nonzero lattice vectors into the shells $nL \leq |\lambda| < (n+1)L$, $n \geq 1$. Disks of radius $L/2$ about distinct lattice points have disjoint interiors. A shell is contained in the disk of radius $(n+3/2)L$ after adjoining these smaller disks, so its number of points is at most $(2n+3)^2$. This bound uses only the shortest lattice vector.

For the displacement estimate with $\nu = 1$ and indices $d, e \leq D$, the finite sum in the common jet bound has at most $D+1$ terms. For $r = |\lambda| \geq 20$, each term is at most $D! r^{2D} e^{-r^2/2}$. A row-sum bound on the $(D+1) \times (D+1)$ matrix therefore gives

$$\mathrm{error}_D \leq (D+1)^2 D! \sum_{\lambda \neq 0} |\lambda|^{2D} e^{-|\lambda|^2/2}. \tag{6.58}$$

For $D \le 16$, $r^{2D}e^{-r^2/2}$ decreases on $r \ge 20$. The ratio of consecutive shell majorants is bounded by $2^{33}e^{-600} < 1/2$: the shell-count ratio is less than two, the polynomial ratio is at most $2^{32}$, and the exponential ratio is at most $e^{-600}$. Thus the sum is bounded by twice its first-shell majorant, giving

$$\sum_{\lambda\neq 0} |\lambda|^{2D} e^{-|\lambda|^2/2} \le 50\, 20^{2D} e^{-200}. \tag{6.59}$$

The elementary Taylor bound $e^{25/36} > \sum_{j=0}^{4} (25/36)^j/j! > 2$ proves $\log 2 < 25/36$. It follows that $e^{-200} < 2^{-288}$, $e^{-100} < 2^{-144}$, and $e^{-400} < 2^{-576}$. Applying (6.58) with $D = 16$ to the two-center pair frame, and with $D = 8$ to the raw-jet Gram matrix, yields

$$\epsilon_4 \le \frac{50 \cdot 17^2 \cdot 16! \cdot 20^{32}}{2^{288}} < 2.612 \times 10^{-28}, \tag{6.60}$$

$$e_8 \le \frac{50 \cdot 9^2 \cdot 8! \cdot 20^{16}}{2^{288}} < 2.152 \times 10^{-58}. \tag{6.61}$$

The same shell argument applied directly to the zero-degree pair error and the diagonal charge-$j$ kernels gives

$$\epsilon_2 \le 50\, 2^{-144} < 2.243 \times 10^{-42}, \qquad |k_1(R) - 1| \le 50\, 2^{-288}, \qquad |k_2(R) - 1| \le 50\, 2^{-576}. \tag{6.62}$$

In particular, $k_1$ is positive, as required in (6.1). To keep the final calculation rational and simple, enlarge all five errors $\epsilon_2, \epsilon_4, e_8, |k_1 - 1|, |k_2 - 1|$ to the common allowance $e = 10^{-8}$. Then

$$s \le s_e := 1 + e, \qquad c \le c_e := \frac{1+e}{(1-e)^2}, \qquad \kappa_B \le \kappa_e := 6C^B_{\mathrm{rows}} e. \tag{6.63}$$

Every subtracted term in (6.54)–(6.55) is monotone in these nonnegative errors. Substitute $s_e, c_e, \kappa_e$ and $\epsilon_2 = \epsilon_4 = e$ to obtain rational lower bounds $\underline{A}$ and $\underline{B}$. Their exact values are recorded in Appendix B; direct rational comparison gives

$$A_\Lambda \ge \underline{A} \approx 0.3372416256964786, \qquad \underline{A} > \frac{337241}{10^6} > \frac{1}{3}, \tag{6.64}$$

$$B_\Lambda \ge \underline{B} \approx 0.015928994442432054, \qquad \underline{B} > \frac{15928}{10^6} > 0. \tag{6.65}$$

Equations (6.53), (6.64), and (6.65), together with $K, S_4 \ge 0$, prove $H_B^2 \ge H_B/3$ on every particle sector whenever $\ell(\Lambda) \ge 20$. The threshold is sufficient and deliberately conservative; no optimization of the lattice budget is used.

Finally, take the algebraic direct sum of finite-particle sectors as a common core. On each sector the preceding inequality excludes spectrum in $(0, 1/3)$. The self-adjoint direct sum $H_B = \bigoplus_{N\ge 0} H_{B,N}$ has the same spectral exclusion, and its spectral calculus gives the asserted full-Fock quadratic-form inequality. This step does not require $H_B$ to be a bounded operator on Fock space. At flux $M = 2N$, the common zero-mode argument identifies its two-dimensional Laughlin kernel, so the bound is a gap above that kernel. For a square torus, $\ell(\Lambda) = \sqrt{\pi M}$, and on the even-flux sequence the sufficient condition holds for $N \ge 64$.

# 7 Laughlin zero modes and the geometric scope

**Proposition 7.1** (The zero-energy space at filling $1/m$)**.** *Take $m = 2$ for the bosonic $V_0$ interaction or $m = 3$ for the fermionic $V_1$ interaction. At flux $M = mN$ and any fixed common single-particle twist,* $\dim\ker H = m$. *The kernel is exactly the torus Laughlin space.*

*Proof.* We use the torus Laughlin–Jastrow theta construction of Haldane–Rezayi and Greiter–Schnells–Thomale [5, 3] for the factors and their multipliers. The following argument must additionally identify the kernel of the physical interaction and exhaust it, including the flat twist. This kernel argument is not included in the selected gap-root acceptance. Positivity of (2.3) makes zero energy equivalent to vanishing of the contact pair amplitude at every center, with the other particle coordinates arbitrary. In the bosonic case the wavefunction vanishes on every pair diagonal; exchange symmetry makes its local relative zero at least quadratic. In the fermionic case the wavefunction is already odd in the relative coordinate and the relative-degree-one contact derivative also vanishes, so its zero has order at least three. Analyticity promotes the integral zero condition to the pointwise statement.

Put $E = \mathbb{C}/\Lambda$, choose $y_i = z_i/\omega_1$ and $\tau = \omega_2/\omega_1$, and divide the holomorphic factor of the wavefunction by

$$J_m(y) = \prod_{i<j} \vartheta_1(y_i - y_j \mid \tau)^m. \tag{7.1}$$

The quotient is holomorphic, including at intersections of pair diagonals, because its divisor has no negative component. The original line bundle on $E^N$ has Chern matrix $MI$. The Jastrow factor has matrix $m(NI - \mathbf{1}\mathbf{1}^{\mathsf{T}})$. At $M = mN$ the quotient therefore has matrix

$$mNI - m(NI - \mathbf{1}\mathbf{1}^{\mathsf{T}}) = m\mathbf{1}\mathbf{1}^{\mathsf{T}}. \tag{7.2}$$

The flat part also descends through the sum map, not just the Chern class. In a theta-function gauge the original holomorphic factor obeys $f(y_i + 1) = e^{i\phi_1} f$ and $f(y_i + \tau) = e^{i\phi_2} e^{-i\pi M(2y_i+\tau)} f$. The theta multipliers show that, after division, a shift of any one coordinate has multiplier

$$(-1)^{m(N-1)} e^{i\phi_1}, \qquad (-1)^{m(N-1)} e^{i\phi_2} e^{-i\pi m(2Z+\tau)}, \qquad Z = \sum_i y_i. \tag{7.3}$$

Hence the quotient bundle is exactly $\Sigma^* L$, where $\Sigma : E^N \to E$ is summation and $L$ is a degree-$m$ line bundle with the indicated fixed flat twist. The fibers are connected compact copies of $E^{N-1}$. A holomorphic section of a pullback is constant along each fiber in a local trivialization, so

$$H^0(E^N, \Sigma^* L) \cong H^0(E, L), \qquad \dim H^0(E, L) = m. \tag{7.4}$$

The last equality is Riemann–Roch on the elliptic curve for positive degree. Conversely, every such COM theta section multiplied by $J_m$ has the required statistics, boundary conditions, and pair zeros, so it is a zero mode. This proves both existence and exhaustion. The general torus theta construction is consistent with [5, 3]; the rectangular-torus pairing formulas and the fermionic threefold exhaustion statement in [10] provide complementary background, not the proof for arbitrary shape and twist used here. □

For a nonnegative self-adjoint operator, $H^2 \geq \gamma H$ is equivalent by spectral calculus to $\mathrm{spec}(H) \subset \{0\} \cup [\gamma, \infty)$, or $H \geq \gamma(I - P_0)$ with $P_0$ the full kernel projection. This needs no uniqueness of the ground state. Applying Proposition 7.1 proves Corollary 2.3. The torus relations are $M = 2N$ and $M = 3N$, not the spherical shifted relations $2(N-1)$ and $3(N-1)$.

At fixed shape the dimensionless systole grows as $\sqrt{M}$, uniformly on compact shape families. More general families are covered whenever their actual systole tends to infinity, even if their aspect ratio is not bounded. Conversely, a long displayed basis vector does not compensate for a short nonzero lattice combination. The proofs do not address fixed short circumferences. No claim uniform over arbitrary degenerating torus shapes follows from these estimates.

# 8 Discussion

The two proofs identify a common geometric mechanism with different statistical costs. Magnetic periodization preserves finite positive relative operators exactly; it does not preserve products of arbitrary

plane operators. The missing product information is recovered through the full-COM pair frame and the explicit connected-kernel unfolding. This separation explains why local few-body sum-of-squares data can lead to particle-number-uniform torus estimates without a semidefinite program whose dimension grows with flux.

The bosonic comparison is not obtained by changing signs in the fermionic proof. Its coherent thermal operators first dominate relative cutoffs and the negative tail as positive few-body coefficients. Their periodizations become integrals over a torus anchor and unrestricted planar spectator offsets. Occupation inequalities then control those integrals by $H$ and $S_4$. Retaining a positive $S_4$ reserve is what permits arbitrarily large occupations. In contrast, Pauli bounds allow the fermionic finite left factors and tail to be paid for by $H$ alone.

The constants are sufficient lower bounds in a fixed physical normalization, not estimates of the actual excitation energies. The bosonic threshold $\ell(\Lambda) \geq 20$ is explicit but intentionally coarse. The fermionic threshold remains existential because the high-degree negative-tail control is taken by dominated convergence. A quantitative treatment of that tail would turn this part into an explicit finite-size statement. Improved planar certificates, including the stronger fermionic bounds in [2], are possible inputs to a further transfer, but their finite matrices and geometric budgets would have to be checked afresh; no such improved torus constant is asserted here.

There are two scope boundaries. First, we do not replace the systole hypothesis by a claim of arbitrary shape uniformity, as discussed in Section 7. Second, this transfer is a quotient construction, not an orbital compression; Appendix C gives an exact disk counterexample and identifies the failing frame estimate. The results also do not prove stability under general perturbations of the torus Hamiltonian. Establishing such stability would require further local-energy and perturbation estimates beyond the finite certificate comparison developed here.

## 9 Conclusion

The bounds below are the mathematical conclusion of the candidate argument, with the formal-verification scope specified in Appendix D. No priority claim is made. Fixed finite Fock-space certificates can be transferred to full bosonic $V_0$ and fermionic $V_1$ interactions on large-systole flat tori by combining positive periodization, parameter COM integration, and all-COM jet-frame bounds. The separate CCR and CAR comparisons give the gap estimates of Theorems 2.1 and 2.2; at $M = 2N$ and $M = 3N$ they apply above the entire two- and three-dimensional Laughlin ground spaces. The proof depends on large actual cycles and on the specified contact normalization, and does not extend by compression to a hard-cutoff disk.

## Contributions and acknowledgments

prepared by Xiang-Jian Hou with advice from Ying-Hai Wu

## A The complete finite fermionic certificate

This appendix specifies all finite input data and all arithmetic required for (5.17) and (5.21). The coefficients originate in [8]; the formulation below uses their planar oscillator values directly. The proof consists of finite sums of integers and rational numbers, followed by exact positivity tests. It neither estimates a many-particle spectrum nor introduces a size-dependent optimization problem.

## A.1 All seven integer rows

Let $P = 10^7$. The first two integers of each row are $t_\rho$ and $\ell_\rho = P\lambda_\rho$. An entry written as $pj : a$ means $a^\rho_{pj} = a$, with $p$ and $j$ read as separate single-digit indices. Indented continuation lines belong to the preceding row. Every omitted entry is zero. Thus the following data, together with (5.14), determine every coefficient without a fitted or rounded decimal input.

```
0 7181518 05:-849651 07:-2958744 15:751659 16:-43084 17:-146216 32:720949 35:-127514
  42:222437 43:-262822 53:-117700 61:-55024 71:-166049
1 -112991 05:142772 06:-142342 07:527055 08:50742 16:197917 18:-18343 24:5764 27:-21183
  32:75832 35:-6115 45:2660 53:-9455 61:-27091 63:-3494 71:479 72:-15331
1 -4247311 05:-12207250 06:-14064191 07:-14063663 08:-15662679 16:483831 18:3987214
  24:-814558 27:888887 32:966462 35:145978 45:-171722 53:34350 61:389319 63:97603
  71:412447 72:-53786
2 3438746 14:6689266 15:6620205 16:6625065 17:4969272 18:4878238 27:-1387380 36:-586188
  37:344087 42:-1120634 45:-318530 46:559169 55:157455 61:1207103 63:211617 64:33240
  71:577001 73:-142578
5 -98645 05:-2719776 08:-8519913 14:-949523 17:-1952105 18:-1562606 28:-8919966 32:248826
  35:-1017623 37:3656005 38:-2081871 42:-33163 43:-305905 46:35677 47:-1320749
  48:-1040378 53:502094 55:1060633 58:-998003 61:-113617 63:184919 64:209239 65:-68198
  67:-266297 72:-606696 73:-563620 74:437345 76:-432189
5 -3160276 05:-1698129 08:10575875 14:-252576 17:-8002616 18:22592844 28:7370640
  32:289294 35:1400647 37:1987740 38:4030699 42:-1731401 43:-1034062 46:-1106371
  47:-1257051 48:1461735 53:1408149 55:1136847 58:1878501 61:-2838516 63:-734099
  64:1062416 65:11745 67:611312 72:-480776 73:20459 74:316925 76:37092
7 1441784 07:3441286 08:-6097155 37:-1002399 47:-251321 58:1143225 66:-17397 71:-1007464
  73:-662286 76:-344621 78:-1115284
```

The row support satisfies $0 \le p \le 7$, $0 \le j \le 8$, and $0 \le p + j - t_\rho \le 8$. Direct integer addition gives

$$\sum_{\rho=1}^{7} \ell_\rho^2 = 93527408868499, \qquad \eta = \frac{93527408868499}{10^{14}}. \tag{A.1}$$

## A.2 Exact three-body arithmetic

For $c \in \{1, 2\}$ and integers $0 \le z, p \le T$, define the integer polynomial coefficient

$$\mathsf{U}(c, z, T, p) = \sum_{h=\max(0,p+z-T)}^{\min(p,z)} (-1)^{z-h} \binom{z}{h} \binom{T-z}{p-h} c^{p-h}. \tag{A.2}$$

Set $\mathsf{U}$ to zero whenever $z$ or $p$ lies outside $[0, T]$. Expanding (5.15) at $u = 1/\sqrt{3}$ gives

$$s^{(1/\sqrt{3})}_{z,T,p} = \mathsf{U}(2, z, T, p) \sqrt{\frac{2^z \binom{T}{z}}{3^T 2^p \binom{T}{p}}}. \tag{A.3}$$

Inserting this expression and (5.14) into (5.16) cancels every square root. Explicitly, let

$$X_{\rho,T}(z) = \sum_{p+j=T} a^\rho_{pj} \mathsf{U}(2, z, T, p). \tag{A.4}$$

Empty sums are zero, and the required rational expression is

$$P^2 e_z = \sum_{\rho=1}^{7} \sum_{T=\max(z,t_\rho)}^{15} \frac{2^z \binom{T}{z}}{3^T 2^{t_\rho} \binom{T}{t_\rho}} X_{\rho,T}(z) \left[ X_{\rho,T}(z) + 2\ell_\rho \mathsf{U}(2, z, T, t_\rho) \right]. \tag{A.5}$$

To see the cancellation directly, for a fixed $T$ the product $\alpha^\rho_{pj} s_{z,T,p}$ equals $a^\rho_{pj}/P$ times a factor independent of $p$. Its square is $2^z \binom{T}{z} / (3^T 2^{t_\rho} \binom{T}{t_\rho})$. The same factor multiplies $\lambda_\rho s_{z,T,t_\rho}$, which proves (A.5).

For each nonzero wedge sector, compute the exact margin

$$m_z = \frac{3}{2}(3z-1)(-1/2)^z - e_z. \tag{A.6}$$

The following table gives the integer floors of $10^6 m_z$. These are obtained by integer division after writing each rational number with a positive denominator.

| $z$ | 2 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|
| $\lfloor 10^6 m_z \rfloor$ | 1298313 | 211317 | 249 | 250 | 249 | 250 | 250 |

| $z$ | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|
| $\lfloor 10^6 m_z \rfloor$ | 250 | 249 | 250 | 250 | 6470 | 250 |

All thirteen floors are positive, and therefore every $m_z$ is strictly positive. This proves (5.17) without floating-point sign decisions. The sectors $z = 0, 1, 3$ are absent because their wedge maps vanish identically.

For completeness, the constant controlling the torus Gram error is equally rational. Put $A_{\rho,i} = \sum_{p+j-t_\rho=i} (\alpha^\rho_{pj})^2$. Squaring (5.14) gives

$$A_{\rho,i} = \frac{1}{P^2} \sum_{p+j-t_\rho=i} (a^\rho_{pj})^2 \frac{2^{p-t_\rho} t_\rho! i!}{p! j!}. \tag{A.7}$$

The nonzero-entry counts $n_\rho$, maximizing indices $i_\rho$, and maxima are

| $\rho$ | $n_\rho$ | $i_\rho$ | $A_{\rho,i_\rho}$ |
|---|---|---|---|
| 1 | 12 | 8 | 237052991863/390625000000 |
| 2 | 16 | 6 | 1154980099359/1400000000000000 |
| 3 | 16 | 8 | 4366100262613/15000000000000 |
| 4 | 17 | 5 | 90518611137713/600000000000000 |
| 5 | 27 | 5 | 289563620848499/5600000000000000 |
| 6 | 27 | 2 | 9393778357061/150000000000000 |
| 7 | 10 | 8 | 77741150041/6250000000000 |

Comparison with the other eight rational values in each row verifies the maxima. Multiplying by $n_\rho$ and summing gives

$$C^{\mathrm{F}}_\alpha = \sum_{\rho=1}^{7} n_\rho A_{\rho,i_\rho} = \frac{297842353828763087}{16800000000000000}. \tag{A.8}$$

### A.3 Rational reconstruction of all four-body blocks

Fix $1 \le D \le 23$ and let all copy indices range over $\mathcal{R}_D$. Define

$$u_r = \frac{1}{r!(D-r)!}, \qquad (E_D)_{rs} = \sqrt{u_r u_s}\, Y_{rs}, \qquad (G_D)_{rs} = \sqrt{u_r u_s}\, Z_{rs}. \tag{A.9}$$

We now express $Y$ and $Z$ entirely over the rationals. First put

$$\mathsf{V}_r(D,T;p,j,k) = \mathsf{U}(1,r,j+k,j)\mathsf{U}(1,D-r,T-r,p), \tag{A.10}$$

with value zero for $r > j+k$. Two expansions of (5.15) at $u = 1/\sqrt{2}$ give

$$S_r(D,T;p,j,k) = \mathsf{V}_r(D,T;p,j,k)\sqrt{2^{1-(j+k)-T+r}\frac{j!k!p!\,u_r}{(T-D)!}}. \tag{A.11}$$

Consequently, the matrix defined by (5.20) is determined by

$$P^2 Y_{rs} = -\sum_{\rho=1}^{7} \sum_{\substack{(p,j),(q,l)\in \mathrm{supp}(a^\rho) \\ T\geq D}} a^\rho_{pj} a^\rho_{ql} \mathsf{V}_r(D,T;p,j,k)\mathsf{V}_s(D,T;q,l,i) \\ \times 2^{1-2T+p+q-t_\rho+(r+s)/2} \frac{i!k!t_\rho!}{(T-D)!}, \tag{A.12}$$

where $i = p+j-t_\rho$, $k = q+l-t_\rho$, and $T = p+j+k$. Both sums over input entries are ordered. To verify the square-root cancellation, multiply the two factors in (A.11) by the two coefficient factors in (5.14). The factorial product under the combined radical is $(t_\rho!i!k!)^2 u_r u_s/((T-D)!)^2$. Since $j+k = T-p$ and $l+i = T-q$, the exponent of two under that radical is $2+2p+2q-2t_\rho-4T+r+s$. Taking its positive square root and removing $\sqrt{u_r u_s}$ gives exactly (A.12).

To reconstruct the Gram matrix, let $A$ be an increasing quadruple of nonnegative indices with sum $D+1$, and put $f(A) = \prod_{b\in A} b!$. For a choice $x<y$ from $A$, write the complementary indices as $j<k$ and set $p = x+y-1$. Let $\sigma(x,y,j,k)$ be the sign of sorting this ordered list into increasing order. Define

$$L_r(A) = 2^{(1-2D+r)/2} \sum_{\substack{x,y\in A,\ x<y \\ \{j,k\}=A\setminus\{x,y\},\ j<k}} \sigma(x,y,j,k)(x-y)p!j!k!\mathsf{V}_r(D,D;p,j,k). \tag{A.13}$$

Indeed, the coefficient of $e_x \wedge e_y$ in $v_p$ is $(x-y)\sqrt{p!/(2^p x!y!)}$ when $x+y = p+1$. Multiplication by (A.11), followed by the wedge sorting sign, shows that the coefficient of $w_{D,r}$ on the normalized Slater wedge $A$ is $\sqrt{u_r/f(A)}L_r(A)$. Taking inner products in this orthonormal wedge basis therefore gives

$$Z_{rs} = \sum_{\substack{A\subset\mathbb{Z}_{\geq 0},\ |A|=4 \\ \sum_{b\in A} b = D+1}} \frac{L_r(A)L_s(A)}{f(A)}. \tag{A.14}$$

All powers of two in (A.12) and (A.13) have integer exponents because $r, s$ are odd. Thus $Y$ and $Z$ are rational matrices. Empty quadruple sums, which occur at small $D$, are exactly zero.

Write $D_u = \mathrm{diag}(u_r)$ and $D_{\sqrt{u}} = \mathrm{diag}(\sqrt{u_r})$. Introduce rational matrices

$$\mathsf{B} = \mathrm{diag}((2\mathbf{1}_{r=1} + \varepsilon_{\mathrm{F}})u_r) - D_u Y D_u, \qquad \mathsf{M} = Z\mathsf{B}Z. \tag{A.15}$$

Here $\mathsf{M}$ is a certificate matrix, distinct from the magnetic flux $M$. Substitution into (A.9) gives the exact congruence

$$G_D(2\Pi_1 - E_D + \varepsilon_{\mathrm{F}} I)G_D = D_{\sqrt{u}}\mathsf{M}D_{\sqrt{u}}. \tag{A.16}$$

Every $u_r$ is strictly positive. Hence proving $\mathsf{M} \geq 0$ proves the desired certificate. This transformation only uses a positive diagonal congruence; it does not invert $G_D$ or discard its kernel.

## A.4 Exact positive-semidefiniteness tests

The finite matrices in (A.15) can be tested by symmetric Schur elimination using rational numbers throughout. Order the copy indices increasingly and initialize $A = \mathsf{M}$. At each remaining index, read its diagonal entry $a$. If $a<0$, the matrix is not positive semidefinite. If $a = 0$, every other entry in that row must be exactly zero; otherwise the associated $2\times 2$ principal minor has determinant $-|b|^2 < 0$. A zero row and column may be removed without changing the rank. If $a>0$, write the remaining matrix as

$$A = \begin{pmatrix} a & b^\dagger \\ b & C \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ b/a & I \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & C - bb^\dagger/a \end{pmatrix} \begin{pmatrix} 1 & b^\dagger/a \\ 0 & I \end{pmatrix}. \tag{A.17}$$

This invertible congruence reduces positivity to that of the smaller Schur complement $C - bb^\dagger/a$ and increases the accumulated rank by one. Iterating proves positive semidefiniteness if every pivot is positive

or has an exactly zero remaining row. Rational subtraction and division determine the decision without a tolerance for small eigenvalues.

A second finite check can be obtained from characteristic coefficients. For a block of dimension $d = |\mathcal{R}_D|$, set $J_0 = I_d$ and recursively compute

$$X_b = \mathsf{M} J_{b-1}, \qquad c_b = \frac{\operatorname{Tr} X_b}{b}, \qquad J_b = c_b I_d - X_b, \qquad 1 \le b \le d. \tag{A.18}$$

Newton's identities, or direct coefficient comparison in the polynomial adjugate identity, give

$$\det(xI_d + \mathsf{M}) = x^d + c_1 x^{d-1} + \cdots + c_d. \tag{A.19}$$

For example, the recursion starts with $c_1 = \operatorname{Tr}\mathsf{M}$ and $c_2 = ((\operatorname{Tr}\mathsf{M})^2 - \operatorname{Tr}\mathsf{M}^2)/2$. If all $c_b \ge 0$, the polynomial in (A.19) is positive for every $x > 0$. Since $\mathsf{M}$ is real symmetric, a negative eigenvalue would give a positive root, a contradiction. Thus nonnegative characteristic coefficients are an independent exact criterion for this calculation.

Evaluating the finite sums (A.2), (A.12), (A.13), and (A.14) at $\varepsilon_{\mathrm{F}} = 3/10^6$ gives the following results. The rank is the number of positive Schur pivots. The sign string lists $c_1, \dots, c_d$ in order; a plus means a strictly positive rational numerator with positive denominator, and zero means exact zero. Every block passes the Schur test, including its zero-row checks, and all characteristic coefficients are nonnegative.

| $D$ | $d$ | rank($\mathsf{M}$) | Signs |
|---|---|---|---|
| 1 | 1 | 0 | `0` |
| 2 | 1 | 0 | `0` |
| 3 | 2 | 0 | `00` |
| 4 | 2 | 0 | `00` |
| 5 | 3 | 1 | `+00` |
| 6 | 3 | 0 | `000` |
| 7 | 4 | 1 | `+000` |
| 8 | 4 | 1 | `+000` |
| 9 | 5 | 2 | `++000` |
| 10 | 5 | 1 | `+0000` |
| 11 | 6 | 3 | `+++000` |
| 12 | 6 | 2 | `++0000` |
| 13 | 7 | 4 | `++++000` |
| 14 | 7 | 3 | `+++0000` |
| 15 | 8 | 5 | `+++++000` |
| 16 | 8 | 4 | `++++0000` |
| 17 | 9 | 6 | `++++++000` |
| 18 | 9 | 5 | `+++++0000` |
| 19 | 10 | 7 | `+++++++000` |
| 20 | 10 | 6 | `++++++0000` |
| 21 | 11 | 8 | `++++++++000` |
| 22 | 11 | 7 | `+++++++0000` |
| 23 | 12 | 9 | `+++++++++000` |

The matrices have dimension at most twelve, and every sum ranges over the printed integer rows and explicitly bounded indices. Equations (A.16) and (A.17), or alternatively (A.19), therefore prove all twenty-three instances of (5.21). Their kernels are retained throughout.

Finally, the finite arithmetic and the analytic tail bound give the exact limiting budget

$$1 - \eta - \frac{61}{4096} - 1222\varepsilon_{\mathrm{F}} = \frac{4616733319001}{10^{14}} > 0. \tag{A.20}$$

The extension from these finite matrices to the full torus Hamiltonian is supplied by the positive congruence and the $H$-relative estimates in Section 5; finite arithmetic alone is not a substitute for those analytic steps.

# B Exact definition and verification of the bosonic finite certificate

This appendix specifies the finite arithmetic input to (6.40)–(6.41). The construction starts from rational row data, forms the physical normal-ordered coefficients, takes a finite center-of-mass partial trace, and tests twenty-six rational quadratic forms. Each step is stated explicitly so that the finite verification can be reproduced independently of the implementation that generated the rows. The analytic transfer to the torus and the occupation estimates are proved in Section 6; they are not inferred from these finite tests.

## B.1 Frozen rows and canonical polynomial coordinates

The numerical input consists of the integers in the accompanying file `data/bosonic/exact_shifted_certificate_rows.json`, taken from the fixed source accompanying Shen [15]. No fitted coefficient is recomputed for the torus. For channel $t = 0, 1, 2, 3$, the ordered list of columns is

$$\mathcal{J}_t = (\emptyset) \,\|\, \big((p, d-p) : d = t, t+1, \dots, 8;\ p = 0, 1, \dots, d\big). \tag{B.1}$$

Here $\emptyset$ is the first column, for the pair term, and $\|$ denotes concatenation. The file stores this list as `terms`, the rows as `integer_scaled_rows`, and the common denominator as `common_denominator`. The complete sizes are

| $t$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $\lvert\mathcal{J}_t\rvert$ | 46 | 45 | 43 | 40 |
| $\#\{\rho : t_\rho = t\}$ | 31 | 37 | 36 | 34 |

, denominator $= 10^{12}$. (B.2)

Dividing the stored integer $m_{\rho,a}$ by $10^{12}$ defines $y_{\rho,a}$, including $y_{\rho,\emptyset} = y_0^\rho$. Equation (6.36), with the positive square root for each $\kappa_p$, uniquely decodes the physical real coefficients. Equivalently, the row itself is the rational polynomial operator (6.37). These specifications fix the input without any floating-point rounding.

For $k$ particles and total orbital degree $d$, use the lexicographically ordered basis

$$\mathcal{B}_{k,d} = \{A = (a_1, \dots, a_k) : 0 \le a_1 \le \cdots \le a_k,\ \textstyle\sum_\ell a_\ell = d\}, \qquad |A]_x = \prod_{j\ge 0} x_j^{n_j(A)} |0\rangle. \tag{B.3}$$

The occupation $n_j(A)$ counts occurrences of $j$ in $A$. The Gram matrix in this basis is diagonal:

$$G_{k,d} = \operatorname{diag}(g_A : A \in \mathcal{B}_{k,d}), \qquad g_A = \prod_{j\ge 0} n_j(A)!\,(j!)^{n_j(A)}. \tag{B.4}$$

A coefficient matrix acts on column vectors. If $O$ is an operator on this degree block, its quadratic-form matrix is $G_{k,d}O$. Physical adjoints obey $\partial_j^* = x_j/j!$ and $x_j^* = j!\partial_j$, rather than $\partial_j^* = x_j$. For a map $F$ between two blocks, its adjoint matrix is $G_{\text{in}}^{-1} F^{\mathsf{T}} G_{\text{out}}$. Thus all adjoints and all form matrices can be computed over the rational numbers.

For completeness, every monomial matrix entry is fixed by the elementary action

$$x^\mu \partial^\nu \prod_j x_j^{n_j} = \left[\prod_j (n_j)_{\nu_j}\right] \prod_j x_j^{n_j - \nu_j + \mu_j}, \qquad (n)_r = \begin{cases} n!/(n-r)!, & n \ge r, \\ 0, & n < r. \end{cases} \tag{B.5}$$

The coefficient is zero whenever a required annihilation is unavailable. Ordered pair sums are retained as ordered sums, including both terms when the two orbital labels differ; this is required by $\widehat{B}_p = \sum_{a+b=p} \partial_a \partial_b$. No additional factorial is attached to the physical lift.

## B.2 Direct rational construction of the body coefficients

It is sufficient to form the rational row Gram matrices

$$\Gamma^{(t)}_{ab} = \sum_{\rho: t_\rho = t} y_{\rho,a} y_{\rho,b} = 10^{-24} \sum_{\rho: t_\rho = t} m_{\rho,a} m_{\rho,b}, \qquad a, b \in \mathcal{I}_t. \tag{B.6}$$

Write $P_p = \widehat{B}^*_p = \sum_{a+b=p} x_a x_b/(a!b!)$. For a noninitial column $a = (p, j)$ set $i = p + j - t$, and for $b = (p', j')$ set $i' = p' + j' - t$. Directly commuting $\partial_i$ through $x_{i'}$ gives the following rational normal-ordered operators:

$$\begin{aligned} \mathsf{M}_3 = &\sum_{t=0}^{3} \sum_{a=(p,j)\in\mathcal{I}_t\setminus\{\emptyset\}} \Gamma^{(t)}_{\emptyset a} \left( P_t x_i \partial_j \widehat{B}_p + (P_t x_i \partial_j \widehat{B}_p)^* \right) \\ &+ \sum_{t=0}^{3} \sum_{a=(p,j),\, b=(p',j')} \Gamma^{(t)}_{ab} \frac{i!}{j!} \delta_{ii'} P_p x_j \partial_{j'} \widehat{B}_{p'}, \end{aligned} \tag{B.7}$$

$$\mathsf{M}_4 = \sum_{t=0}^{3} \sum_{a=(p,j),\, b=(p',j')} \Gamma^{(t)}_{ab} \frac{i!}{j!} P_p x_j x_{i'} \partial_i \partial_{j'} \widehat{B}_{p'}. \tag{B.8}$$

The double sums run over noninitial columns of $\mathcal{I}_t$. For example, $(x_i \partial_j \widehat{B}_p)^* = (i!/j!) P_p x_j \partial_i$, which explains the factorial ratio in both formulas. These equations are precisely (6.38)–(6.39) after the change of variables, and they need no recursive subtraction between particle sectors. Each monomial in $\mathsf{M}_k$ has exactly $k$ creators and $k$ annihilators. Restricting it to $k$ particles therefore gives its physical $k$-body coefficient itself.

Let $M_{k,d}$ be the matrix obtained from (B.7) or (B.8) on $\mathcal{B}_{k,d}$ using (B.5). Its form matrix is $G_{k,d} M_{k,d}$. The three-body matrix vanishes for $d > 8$ and the four-body matrix vanishes for $d > 16$, because the creators' total orbital degrees in the displayed monomials obey these bounds. The two-body row contribution is fixed separately by

$$\eta = \sum_{t=0}^{3} \Gamma^{(t)}_{\emptyset\emptyset} \frac{2^{t+1}}{t!}. \tag{B.9}$$

The planar target matrices have equally explicit rational definitions. On $\mathcal{B}_{k,d}$, let $H^{\mathrm{pl}}_{k,d}$ be the matrix of

$$H^{\mathrm{pl}}_k = \sum_{p\geq 0} \frac{p!}{2^{p+1}} P_p \widehat{B}_p. \tag{B.10}$$

Only $p \leq d$ contribute. For three particles define $S^{\mathrm{pl}}_{3,d} = (H^{\mathrm{pl}}_{3,d})^2 - H^{\mathrm{pl}}_{3,d}$. This is the connected coefficient because the planar pair operator is a projection and no four-body term acts on three particles. For four particles the zero-contraction coefficient on degree $d$ is the matrix of

$$S^{\mathrm{pl}}_{4,d} = \sum_{p+q=d} \frac{p!q!}{2^{d+2}} P_p P_q \widehat{B}_p \widehat{B}_q \Big|_{\mathcal{B}_{4,d}}. \tag{B.11}$$

Here $p + q = d$ because the four annihilators leave the vacuum. The ordered sum over $p, q$ is the zero-contraction sum from the square of $\sum_p B^\dagger_p B_p$, so its normalization agrees with the physical $S_4$.

## B.3 Relative blocks and the exact center-of-mass trace

On the $k$-particle space, the physical oscillator operators become the integer-coefficient polynomial operators

$$L_{\mathrm{osc}} = \sum_{j\geq 1} j x_{j-1} \partial_j, \qquad L^\dagger_{\mathrm{osc}} = \sum_{j\geq 0} x_{j+1} \partial_j, \qquad [L_{\mathrm{osc}}, L^\dagger_{\mathrm{osc}}] = kI. \tag{B.12}$$

For each $z$, form the matrix of $L_{\mathrm{osc}} : \mathcal{B}_{k,z} \to \mathcal{B}_{k,z-1}$ by (B.5). A rational column matrix $U_{k,z}$ spanning its kernel defines the full relative space. A deterministic choice is the usual reduced-row-echelon kernel basis: scan columns in the order (B.3), use the first available pivot, and list the free-variable vectors in increasing column order. No orthogonalization is needed.

Write $D^{(m)}_{k,z} = (L^{\dagger}_{\mathrm{osc}})^{m-z} U_{k,z}$ for the unnormalized descendants in total degree $m \geq z$. The oscillator relation implies that the normalized descendant is $D^{(m)}_{k,z}/\sqrt{k^{m-z}(m-z)!}$; the relative basis itself may remain nonorthonormal. Let $d_3 = 8$ and $d_4 = 16$. The relative form matrix of the averaged row coefficient is then

$$\overline{M}_{k,z} = \frac{2}{k} \sum_{m=z}^{d_k} \frac{(D^{(m)}_{k,z})^{\mathsf{T}} G_{k,m} M_{k,m} D^{(m)}_{k,z}}{k^{m-z}(m-z)!}. \tag{B.13}$$

This is the full center-of-mass partial trace, with the same factor $2/k$ as (6.10). Every center-of-mass descendant with nonzero anchored coefficient is included. The sum is finite because the anchored degree support is finite. The square-root descendant normalization has canceled between the two sides of the quadratic form, so every matrix entry remains rational.

The relative target forms are

$$h_{k,z} = U^{\mathsf{T}}_{k,z} G_{k,z} H^{\mathrm{pl}}_{k,z} U_{k,z}, \qquad s_{k,z} = U^{\mathsf{T}}_{k,z} G_{k,z} S^{\mathrm{pl}}_{k,z} U_{k,z}. \tag{B.14}$$

Thus the twenty-six matrices that must be positive are exactly

$$E_{3,z} = s_{3,z} - \overline{M}_{3,z} + \delta_B h_{3,z}, \qquad z = 0, \dots, 8, \tag{B.15}$$

$$E_{4,z} = \alpha_B s_{4,z} - \overline{M}_{4,z} + \delta_B h_{4,z}, \qquad z = 0, \dots, 16. \tag{B.16}$$

All symbols on the right-hand sides have been defined by finite sums, rational arithmetic, and the fixed integer input. Positive semidefiniteness in this nonorthogonal relative basis is equivalent to the physical operator inequalities, because $U_{k,z}$ is an injective parametrization of the entire relative space.

## B.4 Exact positivity, ranks, and zero kernels

Each form matrix is first checked to be symmetric over the rational numbers. The positivity test repeatedly chooses a strictly positive diagonal entry $p$, permutes it to the first position, and replaces the remaining block $C$ by its Schur complement $C - bb^{\mathsf{T}}/p$. The justification is the exact congruence

$$\begin{pmatrix} p & b^{\mathsf{T}} \\ b & C \end{pmatrix} \sim \begin{pmatrix} p & 0 \\ 0 & C - bb^{\mathsf{T}}/p \end{pmatrix}. \tag{B.17}$$

At every stage, all diagonal entries are required to be nonnegative. If no positive diagonal entry remains, every remaining entry must be exactly zero. This last check is essential: a nonzero off-diagonal entry in a matrix with zero diagonal would prevent positive semidefiniteness. Termination with positive pivots and an exact zero block proves positive semidefiniteness by the displayed congruence, and the number of positive pivots is its rank. No numerical eigenvalue threshold is used.

For each block, rational row reduction gives $\ker h_{k,z}$. Verify exactly that $E_{k,z} \ker h_{k,z} = 0$. Independently apply the same positive-pivot test to $h_{k,z}$ and compare ranks. The resulting dimensions and ranks of all blocks are as follows; each rank shown is simultaneously the rank of $E_{k,z}$ and $h_{k,z}$.

| $k=3,\ z$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| dim | 1 | 0 | 1 | 1 | 1 | 1 | 2 | 1 | 2 |
| rank | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

(B.18)

| $k=4,\ z$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| dim | 1 | 0 | 1 | 1 | 2 | 1 | 3 | 2 | 4 |
| rank | 1 | 0 | 1 | 1 | 2 | 1 | 3 | 2 | 4 |

(B.19)

| $k=4,\ z$ | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|
| dim | 3 | 5 | 4 | 7 | 5 | 8 | 7 | 10 |
| rank | 3 | 5 | 4 | 6 | 5 | 7 | 6 | 8 |

(B.20)

The input includes both degree-one zero-dimensional blocks. Across the other blocks the procedure produces sixty-seven strictly positive rational pivots; the remaining Schur blocks are identically zero. The full fractions are included with the machine-readable certificate results. The zero-kernel checks and equal ranks show that each $E_{k,z}$ is strictly positive on the physical image of $H^{\mathrm{pl}}_{k,z}$ and vanishes on its kernel. In particular, this proves (6.40)–(6.41) as positive coefficient inequalities on the full relative multiplicity spaces.

The row constant used for the torus commutator error is also obtained directly from the same integers:

$$C^B_{\mathrm{rows}} = \sum_{t=0}^{3} \sum_{\rho\cdot t_\rho = t} \sum_{(p,j)\in \mathcal{J}_t\setminus\{\emptyset\}} (y_{\rho,pj})^2 \frac{2^{p+1}(p+j-t)!}{p!j!}. \tag{B.21}$$

Equations (B.9) and (B.21) give exactly the fractions in (6.42) and (6.46). Thus the positivity certificate and the cost of its noncanonical torus commutators are computed from one and the same fixed set of physical row coefficients.

### B.5 Exact scalar arithmetic for the explicit threshold

The scalar budget uses only rational numbers after the shell estimates have replaced exponentials by powers of two. With $e = 1/10^8$, $s_e = 1+e$, $c_e = (1+e)/(1-e)^2$, and $\kappa_e = 6C^B_{\mathrm{rows}}e$, define

$$\underline{A} = 1 - e - \eta - s_e[c_9 + (\delta_B + \kappa_e)C^B_3] - \delta_B C^B_4 s_e^2, \tag{B.22}$$

$$\underline{B} = 1 - \alpha_B(1+e) - s_e c_e[c_9 + (\delta_B + \kappa_e)C^B_3] - 2\delta_B C^B_4 s_e^2 c_e. \tag{B.23}$$

Their exact values are

$$\underline{A} = \frac{8183525179652905540939777339971061478405006776240354085927}{24266058979973529600000000000000000000000000000000000000000000000}, \tag{B.24}$$

$$\underline{B} = \frac{193266955450524232405968648894820259194606401114955533998034101 51}{12133029247326176213567652998676480000000000000000000000000000000000}. \tag{B.25}$$

Multiplication by the positive denominators verifies $\underline{A} > 337241/10^6 > 1/3$ and $\underline{B} > 15928/10^6 > 0$. If decimal lower bounds are desired, valid downward truncations are $\underline{A} \geq 0.3372416256964785$ and $\underline{B} \geq 0.015928994442432054$. Rounded decimal displays are approximations only; the exact fractions, rather than floating-point rounding, determine the signs in the proof.

For reference, the rational shell bounds used before the common allowance are

$$b_4 = \frac{50\cdot 17^2\cdot 16!\cdot 20^{32}}{2^{288}}, \qquad b_8 = \frac{50\cdot 9^2\cdot 8!\cdot 20^{16}}{2^{288}}, \qquad b_2 = \frac{50}{2^{144}}. \tag{B.26}$$

Each is strictly less than $1/10^8$, as are $50/2^{288}$ and $50/2^{576}$ for the diagonal kernels. Consequently, the common substitution in (B.22)–(B.23) enlarges every error in the safe direction. This finite arithmetic closes the analytic estimate; it does not replace the positive thermal transfer, the connected-kernel identity, or the all-center frame bounds used to obtain that estimate.

## C Why hard orbital truncation does not follow from the transfer

A hard disk cutoff is not a magnetic-periodic quotient. Let $P_m$ retain the planar orbitals $e_0, \dots, e_m$ and compress the fermionic relative-degree-one projector to $\bigwedge^2 P_m U$. The highest total-degree sector $2m-1$ contains only $e_{m-1}\wedge e_m$. The pair coefficient (2.2) therefore gives its exact eigenvalue

$$E_m = \frac{(2m-2)!}{2^{2m-2}(m-1)!m!} = \frac{1}{m}\frac{\binom{2m-2}{m-1}}{4^{m-1}} \sim \frac{1}{\sqrt{\pi}m^{3/2}} > 0. \tag{C.1}$$

For example, $E_{12} = 29393/2097152$. Hence a particle-number and cutoff-uniform positive gap for all these disk compressions is false already with two particles. Compression does not preserve a squared lower bound: for any many-body projection $P$,

$$(PHP)^2 = PH^2P - PH(I-P)HP. \tag{C.2}$$

The omitted term is nonnegative and cannot be discarded in a lower-bound proof. In the present method the exact failure is visible in the pair map: the compressed disk map has a highest-COM singular value squared $E_m$, so its Gram operator is not uniformly close to the identity, even on its nonzero support. Local disk jets also cease to be uniformly close to planar jets outside the droplet, and there is no lattice-image identity (3.7). The full-COM estimate and unfolding, rather than agreement on a few low orbitals, distinguish the torus construction.

# D Reproducibility

The seven fermionic rows come from OpenAI [8, Appendix A]; the 138 bosonic rows come from Shen [15], at the fixed versions linked in the bibliography. The torus argument uses the bosonic source's twenty-six planar blocks, not its 156 finite-sphere interval checks. The retained data and exact constructions are specified in the certificate appendices.

The accompanying Python programs reconstruct the finite coefficient forms and scalar budgets using exact rational arithmetic. The bosonic reconstruction checks all twenty-six planar blocks and their zero kernels; the fermionic reconstruction checks the twenty-three four-body blocks, finite three-body margins, and scalar gap budget. Singular positive semidefinite blocks are tested with exact zero-row and kernel checks, without numerical regularization.

These finite arithmetic checks do not establish the magnetic-periodization identities, the COM trace interchange, the physical normalization correspondence, or the occupation bounds. Those analytic steps are supplied by the candidate arguments in the main text. Selected formal endpoint checks do not constitute formal verification of every theorem and corollary here; the detailed scope is recorded in the accompanying repository's `VERIFICATION.md`.

Finite-size normalization diagnostics gave a fermionic three-body residual about $7.21 \times 10^{-12}$ at $M = 9$, and bosonic positive energies about $0.700809668$ at $(M, N) = (6, 3)$ and $0.586180647$ at $(8, 4)$. These calculations are diagnostics, not inputs to either uniform bound.

Both language editions share equations, constants, assumptions, and labels through one paired-prose source, compiled with XeLaTeX and BibTeX. Build commands, exact data, and executable finite checks accompany the manuscript. AI contributions and human roles are documented in the repository's `AI_CONTRIBUTIONS.md`, following the explicit disclosure practice of Shen [14].

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